MASTER provenance of the symbols of PROJECT spacetimes

Every symbol of the project with one meaning, and the proof that no symbol has two

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1. What this document is

The notebooks, papers and documents on which PROJECT spacetimes rests were written over many years, and they often use the same letter for different objects. The letter g alone stands, in them, for a component of a metric, for a rate of deflation, for a coupling of a self-interaction and for the coupling of a scalar field; the letter M for a mass and for a family of matrices; the letter F for a matrix of numbers and for scattering amplitudes. A reader who meets such a letter cannot know what it means without knowing where the sentence came from.

This document removes that ambiguity for the whole project. It contains

1. the **rules** by which symbols of the project are told apart (Section 2);
2. the list of the **collisions** that were found, with the decision taken for each (Section 3);
3. a **dictionary** from the notation of the author's document on Kohn–Sham matter and gravity on the 7+1 spacetime to the symbols of the project (Section 4);
4. the **proof** that the registry is free of collisions: eight machine checks, with their complete output (Section 5), and the instructions for repeating them (Section 6);
5. the **complete registry**: 246 entries in 24 groups, with 375 names used in programs and data files (Section 7), and the key to the column "origin" (Section 8);
6. the symbols of the **two forms of the self-interaction** on the 4+4 spacetime, form C (the definition in this folder) and form Q (the alternative), in one table with the rule for each (Section 3.2), and the list of the symbols that an earlier version of one document of this folder used and that were **replaced** (Section 3.3).

Every symbol that occurs in this document is defined in Section 7 of this document; nothing has to be looked up elsewhere.

How the registry is used. The registry is one table, `provenance/symbols_master.csv`. A script writes it; a second script checks it and checks the documents and the data files of the folder against it. Every document of the folder `Revision_Deflation_CJT_and_Kohn-Sham_definition_C` uses the symbols of the registry with the meanings given there, and a symbol that is not in the registry may not be introduced without being added to it.

2. The rules

R1. A tag says which spacetime an object belongs to. Objects of the flat 4+4 spacetime carry the tag 1644 (γ^{A}_{1644}, C^{-}_{1644}, W_{1644}), objects of the flat 7+1 spacetime the tag 1671 (γ^{A}_{1671}, χ_{1671}). The fields on the deflating spacetimes carry the tags 1644d and 1671d. Where a section treats one spacetime only, the tag may be dropped; the section says so.

R2. A suffix says which sector an object belongs to. Every quantity of the Vorton-Scalar carries the suffix VS: Φ_{VS}, v_{VS}, y_{VS}, λ_{VS}, g_{VS}, α_{VS}, D_{VS}, G_{VS}.

R3. The four kinds of g are written in four different ways.

object	how it is written	example
rate of deflation (the author's names)	upright, two digits, never a subscript	g55, g66, g77
component of the metric of a deflating spacetime	italic, superscript $(4+4)$ or $(7+1)$, two indices	$g_{55}^{(4+4)}$, $g_{\mu\nu}^{(7+1)}$
coupling of the self-interaction	italic, one digit or the letter r as subscript	$g_0, \dots, g_8$, g_r, g_0^{eff}
coupling of the Vorton-Scalar	italic, suffix VS	g_{VS}

R4. Type distinguishes matrices of numbers from amplitudes, and scale factors from indices. Sans-serif type: the Fierz matrix F, the density matrix n, the Hamiltonian matrix h, the two scale factors a, b of the equations of gravity. Italic type: the amplitudes F_S, F_P, F_0, F_1, F_7, F_8; the indices a, b. Script type: the Einstein tensor $\mathcal{G}$, the Lanczos tensor $\mathcal{H}$, the Killing vectors $\mathcal{X}$, the sources $\mathcal{D}, \mathcal{T}, \mathcal{P}, \mathcal{A}$ of the perturbation equations.

R5. An index, an argument or a bar distinguishes objects that share a letter. Λ without index is a momentum cutoff, Λ_8 and $\bar{\Lambda}$ are cosmological terms; κ_8 is the gravitational coupling and $\kappa(k)$ a pair correlation; E_F, E_k, $E_\perp$, E^* are energies and $E^{(k)\mu}{}_\nu$, with the order in parentheses, is a Lovelock tensor; α^j with a superscript are Dirac matrices, $\alpha_1, \alpha_2, \alpha_3$ with a digit as subscript are couplings of the Lovelock terms, α_{VS} and α_c are coupling strengths; σ_8 is the amplitude of clustering, σ_v a velocity dispersion and $\sigma(p)$ a self-energy.

R6. A letter that the author uses keeps the author's meaning, wherever that is possible without a collision: G_S, G_V, V_0, I_r, Θ^K, g_r of his document on Kohn–Sham matter are used here for the same objects. Where a letter of his collides with a symbol of the project, the registry gives the object a new symbol and records his letter (Section 4).

R7. Program names. A name used in a program or as a column of a data file is registered with its symbol. A name may belong to one entry only. Names of columns that are plain words ("content", "case") are listed in the checking script.

3. The collisions that were found, and how each was resolved

3.1 Letters with several meanings

letter	the different objects that used it	resolution in the registry
g	rate of deflation; component of the metric; coupling of the self-interaction; coupling of the Vorton-Scalar	rule R3
g_{1644d}, g_{1671d}	the metric of the first revision on the deflating spacetimes, in which the time-like component was positive on 7+1	retired; the metric is $g^{(4+4)}$, $g^{(7+1)}$ in the author's sign convention (space-like components positive)
η	flat metric; matrix of the squares of the Dirac matrices; function η of the synchronous gauge; sign η_K of the author's document; loading of an evolution	$\eta^{(4+4)}$, $\eta^{(7+1)}$ (metric); η_{1644}, η_{1671} (squares of the matrices); η_{sy}; sgn_K; the word "loading"
F	Fierz matrix F_{rs}; pairing amplitudes F_S, F_P, F_0	$\mathsf{F}_{rr'}$ in sans-serif type for the matrix

letter	the different objects that used it	resolution in the registry
σ	exponent σ_r of the phase of the author's document; amplitude σ_8; velocity dispersion; self-energy $\sigma(p)$	phase ζ_r; σ_8; σ_v; $\sigma(p)$
M	mass of the particles in matter; vertex matrices M_K; gap matrix	mass m^* (for the two halves of the 4+4 field m^*_+, m^*_-); the vertex matrices are written out as Adj Θ^K with the matrix Adj of the adjoint (β on 7+1 and in form Q, C^- in form C); gap matrix M
ρ	number density of matter; energy density of cosmology; coordinate index	number density n and scalar density n_S; ρ is a mass or energy density of cosmology; as an index only as sub- or superscript
Π	interaction part of the pressure	p_{int}
κ	gravitational coupling; pair correlation; ratios of couplings to the number C of the Vorton-Scalar; bound on a cross-section	κ_8; $\kappa(k)$; the ratios are written out, $-G_S/C$, G_V/C, $-g_0/C$, g_1/C, and have no letter of their own; κ_{SI}
Λ	momentum cutoff; range of the contact couplings; cosmological terms	Λ (cutoff); range parameter m_{ct}; Λ_8, $\bar{\Lambda}$
P	polarization of the two halves of the 4+4 field; generator of $U(1)_E$; chiral projectors; momenta of a classical particle	pol_W; P, P_c (generators); $P^{L,R}$; P_μ
a, b	indices of the extra axes; scale factors of the equations of gravity; scale factor of ordinary space	indices a, b; a, b; a_{sp}
s_t	stretch factors of the momenta along the deflating coordinates	written as the ratio b_t^{ref}/b_t
τ	deflation clock; clock of the undisturbed deflation; conformal time; optical depth	τ_d; τ_d^0; τ_{cf}; τ (optical depth, only in the list of cosmological parameters)
K	generator of the evolution on 4+4; index list of the self-interaction; sectional curvature; upper limit of the momentum integral of the gap equation	K, $K_\perp$ (no index or the index $\perp$); K, J as index lists appear only as sub- or superscripts (of Θ, sgn, z, $\mathcal{J}$) and under sums; K_{AB} with two indices; the upper limit is k_{max}
W	the matrix $iC^-\gamma^4$ of the 4+4 spacetime, whose eigenvalues ±1 divide the field into its two halves; kernel of the gap equation; kernel at large momenta	W_{1644}, always with its tag; $W(k,k')$, always with its two arguments; $w_\infty(t)$; as an index, W marks a difference between the two halves: n_W, n_{SW}, pol_W
I	the nine invariants of the self-interaction; the same in the two forms; integrals of the ultraviolet criterion	I_r, $I_0, \dots, I_8$; I_r^{C}, I_r^{Q}; the two integrals are the numbers u_S, u_V
s	sign of the contact couplings in a pair; exponent of the fall of the gap function; mass of the radial mode as an index	s_{ct}; s_Δ; m_s
Ω	spin connection; CJT effective potential; grand potential of one half; energy density of the vacuum of the two halves	Ω_μ (one index); Ω_{CJT}; Ω_{half}; Ω_{vac}
x	coordinates $x0, \dots, x7$; ratio m^*/m; ratio ϕ/v_{VS}	coordinates always with their digit; $x = \phi/v_{VS}$ (equal to m^*/m only without bare mass and contact couplings)
Φ	Vorton-Scalar; state of a pair	Φ_{VS}; Φ, $\Phi(\vec{p})$ without suffix

letter	the different objects that used it	resolution in the registry
Γ	Christoffel symbols; Dirac matrices of the author's 7+1 document; effective action; Euler's gamma function	${\Gamma^{\rho}}_{\mu\nu}$ (three indices); Dirac matrices are γ; $\Gamma[...]$ with its argument in brackets; Γ_{E}
ω	product of the eight Dirac matrices; coefficients of the spin connection; energy of a particle in the author's document; densities of cosmology	ω_{1644}, ω_{1671}; $\omega_{\mu AB}$ (three indices); $E^*(k)$; ω_b, ω_c

3.2 The symbols of the two forms of the self-interaction

On the 4+4 spacetime the self-interaction has two forms in the sector that can be quantized (fields that do not depend on $x5, x6, x7$): **form C**, built with the covariant adjoint $\tilde{\psi} = \psi^\dagger C^-$, which is the definition in this folder, and **form Q**, built with the adjoint $\bar{\psi} = \psi^\dagger \beta$ of the positive Lagrangian, which is the alternative. Every object that exists in both forms carries the letter of its form; an object without such a letter is the same in both. The table collects these symbols; each is an entry of the registry of Section 7.

object	form C	form Q	rule
adjoint of the bilinears	$\tilde{\psi} = \psi^\dagger C^-$	$\bar{\psi} = \psi^\dagger \beta$, $\beta = -i\gamma^4$	a tilde for the covariant adjoint, a bar for the adjoint of the positive Lagrangian; in a formula for both, $\psi^\dagger$Adj
Dirac matrices of the bilinears	γ^A	$\gamma^A_{44q} = -i\gamma^A$ ($A \le 4$), γ^A ($A \ge 5$)	the tag 44q
the 256 matrices, their phases and signs	Θ^K, ζ_r, sgn_K	Θ^K_Q, ζ^Q_r, sgn^Q_K	the letter Q
the nine invariants	I^{C}_r	I^{Q}_r	upright superscript; I_r without superscript is the invariant of the covariant Lagrangian, equal to I^{C}_r
the self-interaction	$\mathcal{L}^{\mathrm{C}}_{\mathrm{self}}$	$\mathcal{L}^{\mathrm{Q}}_{\mathrm{self}}$	upright superscript
tag of a check or of a data file	44c	44q	71s is the self-interaction on 7+1

Objects that are the same in both forms: the matrix W_{1644} and the two halves $\psi_\pm$ of the field ($W_{1644}\psi_\pm = \pm\psi_\pm$); the phase z_K in $C^-\Theta^K = z_K\, W_{1644}\, \beta\, \Theta^K_Q$; the bilinears of the halves $\mathcal{J}^{++}_K$, $\mathcal{J}^{--}_K$, $\mathcal{J}^{+-}_K$, $\mathcal{J}^{-+}_K$; the positive Lagrangian $\mathcal{L}_{\mathrm{kin}} + \mathcal{L}_{\mathrm{mass}}$ and the covariant Lagrangian $\mathcal{L}_{\mathrm{cov}}$; the densities $n_\pm$, $n_{S\pm}$, n_W, n_{SW}, the polarization pol_W, the masses $m^*_\pm$, the energy shifts $V_{0\pm}$, the Fermi momenta $k_{F\pm}$, $k_{F,\mathrm{tot}}$; the four combinations G^{HF}_S, G^{HF}_V, G^{c}_S, G^{c}_V of the effective couplings. The mean-field couplings are named after what they multiply, not after a form: G^{tot}_S, G^{tot}_V (the squares of the total densities), G^{dif}_S, G^{dif}_V (the squares of the differences between the halves), G^{same}_S, G^{same}_V (two particles of the same half), G^{opp}_S, G^{opp}_V (a particle of each half); which of the four combinations each of them equals depends on the form, and the registry gives both assignments.

3.3 Symbols that were used in this folder and were replaced

The first version of the provenance document on the two forms (commit `9f8a206` of the repository) introduced symbols of its own before they had been compared with the registry. Several of them collided, or nearly collided, with registered symbols, and one name of a data column was a symbol of one of the author's notebooks. They were replaced in the next commit, in the documents, in the program and in the data files; check MS8 of Section 5 verifies that none of them occurs in

any document of the folder. A reader who looks at that earlier commit needs this table. (The replaced symbols are printed here as plain program text, so that the check does not find them in this document.)

replaced	by	reason
sans-serif `W` for the kernel of the gap equation, and `W` without tag for the matrix of the two halves	$W(k,k')$ with its two arguments; W_{1644}	one letter for two objects
`T_S`, `T_V`, `D_S`, `D_V`	G_S^{tot}, G_V^{tot}, G_S^{dif}, G_V^{dif} (in programs `G_Stot`, `G_Vtot`, `G_Sdif`, `G_Vdif`)	`T_V` is a symbol of one of the author's notebooks; `D_S` differs only by a capital from the propagator D_s of the radial mode
`a_S`, `a_V`, `b_S`, `b_V`	G_S^{same}, G_V^{same}, G_S^{opp}, G_V^{opp} (in programs `G_Ssame`, `G_Vsame`, `G_Sopp`, `G_Vopp`)	`a_S` differs only by a capital from the curvature a_s of the effective potential of the Vorton-Scalar; a, b are indices and `a`, `b` scale factors
`kappa_0`, `kappa_1` for the ratios of the contact couplings to C	the ratios written out, $-g_0/C$ and g_1/C (the columns of the data files keep the names `kappa_0`, `kappa_1`)	κ_8 is the gravitational coupling
`M_+`, `M_-` for the masses of the two halves	m_+^*, m_-^* (in programs `M_plus`, `M_minus`)	the mass in matter is m^*
`epsilon_K` for the sign of the square of Θ^K	sgn_K, sgn_K^Q	ε is an energy density
`I_r^cov`	I_r^{C}	one way of marking a form
`V_contact`	V_{ct}	the registered symbol
`Omega_1` for the grand potential of one half	Ω_{half}	a digit as index marks other objects (α_1, g_1)
`I_S`, `I_V` for the two integrals of the ultraviolet criterion	u_S, u_V	I_0 is the scalar invariant
sans-serif `w`, `s` of the ultraviolet criterion	$w_\infty(t)$, s_Δ	w is a constant spinor
`k_±`, `V_±`, `w_±`, `s_±` of the two halves	$k_{F\pm}$, $V_{0\pm}$, $E_\pm^*(k)$, $n_{S\pm}^{\mathrm{vac}}$	the registered symbols with the index of the half added
`P_E` for the generator of $U(1)_E$	P (in programs `P`)	the registered symbol
`K_pair` for the curvature of a pair of directions	K_{AB}	the registered symbol
`L_C`, `L_Q`, `L_0`, `L_cov^(0)`, `V_C`	$\mathcal{L}_{\mathrm{self}}^{\mathrm{C}}$, $\mathcal{L}_{\mathrm{self}}^{\mathrm{Q}}$, $\mathcal{L}_{\mathrm{kin}}+\mathcal{L}_{\mathrm{mass}}$, $\mathcal{L}_{\mathrm{cov}}$	the three parts of the Lagrangian have registered names
`K` for the upper limit of a momentum integral; `J`, `R`, `X` and single letters a, b, c for auxiliary quantities	k_{max}; $\mathcal{J}_K^{++}$ and its three companions; $\mathsf{R}_{\mathrm{gap}}$; the quantities written out	K, J are index lists, R is the curvature scalar, X the matrix of the pair along the hidden axis

4. Dictionary: the author's document on Kohn–Sham matter and gravity on 7+1

The author's document that defines, solves and analyzes the 7+1 problem with self-interaction uses its own conventions. The calculations of this project reproduce its numbers; the table says how its notation is translated.

in the author's document	in this project	remark
time coordinate x_0; ordinary space x_1, x_2, x_3; auxiliary space $x_4, \ldots, x_7$	time $x4$; ordinary space $x1, x2, x3$; the four other space-like coordinates $x0, x5, x6, x7$	the exchange of the index values 0 and 4 (swap_{04})

in the author's document	in this project	remark
$\eta = \mathrm{diag}(+1, -1, \dots, -1)$: squares of the Dirac matrices and signs of the metric	η_{1671} with the time at index 4; the metric of the deflating spacetime is $g^{(7+1)}$ with space-like components positive	$\eta^{(7+1)} = -\eta_{1671}$
Dirac matrices Γ_A	γ^A_{1671}	the same real matrices
Christoffel symbols in braces	$\Gamma^\rho{}_{\mu\nu}$	
couplings g_r; invariants I_r; matrices Θ^K, Θ_K	the same symbols	
sign η_K; phase i^{σ_r}	sgn_K; ζ_r	
Fierz matrix F_{rs}	$\mathtt{F}_{rr'}$	
density matrix R	$\mathtt{n}\,\beta$	
Hartree–Fock couplings G_0, G_1	G_S^{HF}, G_V^{HF}	
G_S, G_V, V_0	the same symbols	the mean-field couplings and the vector field
mass in matter M	m^*	the data files keep the name `M`
densities ρ, ρ_S	n, n_S	the data files keep `rho`, `rho_S`
energy of a particle ω	$E^*(k)$	
interaction pressure Π	p_{int}	the data files keep `Pi`
scale factors a, b and rates H_a, H_b	$\mathtt{a}$, $\mathtt{b}$, $H_{\mathtt{a}}$, $H_{\mathtt{b}}$	the data files keep `a`, `b`, `H_a`, `H_b`
gravitational coupling κ	κ_8	
Lovelock tensors $E^{(k)}$, couplings α_k, cosmological term $\bar{\Lambda}$	the same symbols	

5. The proof: eight machine checks

The script `code/scripts/check_symbols.py` reads the registry, every document of the folder and every data file, and verifies eight statements. Its complete output:

```
PROJECT spacetimes, Revision_Deflation_CJT_and_Kohn-Sham_definition_C: checks of the symbols
 ↪  (MASTER registry provenance/symbols_master.csv)
generated by: code/scripts/check_symbols.py
8 of 8 checks passed

[PASS] MS1  the 246 symbol texts of the MASTER registry are pairwise different, and so are its 375
 ↪  names in programs and data files
[PASS] MS2  no definition of the MASTER registry sends the reader to another document (searched
 ↪  for the names of the documents of the project and for the words "main registry", "supplement",
 ↪  "see the")
[PASS] MS3  none of the 66 names in programs that this sub-project introduces itself (entries of
 ↪  the groups 10 to 15 with the origin "new"; names of more than two characters) is a symbol of
 ↪  one of the author's notebooks (1921 symbols compared). Names that are the author's own (G_S,
 ↪  G_V, V_0, p_a, p_b and others of his Kohn-Sham sections) are used for the objects for which he
 ↪  uses them
[PASS] MS4  in the 39 documents of this folder the rates g55, g66, g77 are written upright 619
 ↪  times and never as a letter with a subscript; each of the 79 components of the metric g_{55},
 ↪  g_{66}, g_{77}, g_{tt} carries the superscript (4+4) or (7+1) or stands in a paragraph that
 ↪  names it a component of the metric
[PASS] MS5  the retired symbols do not occur in the documents: \varkappa nowhere; g_{1644d},
 ↪  g_{1671d} only where a paragraph says that they belong to the earlier sign convention
[PASS] MS6  the 14 retired names of programs (tau, kappa_t, Q_over_eps, Omega0, S2, mu and nu as
 ↪  orders of Bessel functions, A, Xi without index, m_b, w0, omega_c for the shell, the metric
 ↪  names g_1644d and g_1671d of the earlier sign convention, and Lambda as the range of the
 ↪  contact couplings) do not occur in the 148 data files that the routines of this folder write
```

```
[PASS] MS7  each of the 1273 columns of the 126 data files of this folder is a registered name, a
 ↪  column of the flat-space part of the project, one of the 273 descriptive words listed in the
 ↪  script, or (213 columns) a systematic name
[PASS] MS8  none of the 15 groups of symbols that the first version of the document on the two
 ↪  forms of the self-interaction used and that collided, or nearly collided, with registered
 ↪  symbols occurs in the 39 documents of this folder (sans-serif W for the kernel of the gap
 ↪  equation -> W(k,k'); T_S, T_V, D_S, D_V -> G_S^tot, G_V^tot, G_S^dif, G_V^dif; a_S, a_V, b_S,
 ↪  b_V -> G_S^same, G_V^same, G_S^opp, G_V^opp; kappa_0, kappa_1 -> the ratios -g_0/C, g_1/C;
 ↪  M_+, M_- for the masses of the two halves -> m*_+, m*_-; epsilon_K for the sign of the square
 ↪  of Theta^K -> sgn_K, sgn_K^Q; I_r^cov -> I_r^C; V_contact -> V_ct; Omega_1 for the grand
 ↪  potential of one half -> Omega_half; I_S, I_V for the two integrals of the ultraviolet
 ↪  criterion -> u_S, u_V; sans-serif w and s of the ultraviolet criterion -> w_infty, s_Delta;
 ↪  k_+-, V_+-, w_+-, s_+- of the two halves -> k_F+-, V_0+-, E*_+-, n_S+-^vac; P_E for the
 ↪  generator of U(1)_E -> P; K_pair -> K_AB; L_C, L_Q, L_0, L_cov^(0), V_C -> L_self^C, L_self^Q,
 ↪  L_kin + L_mass, L_cov)
```

What the checks mean:

- **MS1** No two entries of the registry have the same symbol text, and no name of a program or data file belongs to two entries. This is the statement that no symbol has two meanings.
- **MS2** No definition sends the reader to another document.
- **MS3** No name that this sub-project introduces is already a symbol of one of the author's notebooks with another meaning.
- **MS4** In the documents, a rate is never written as a letter with a subscript, and a component of the metric always carries its superscript.
- **MS5**, **MS6** The retired symbols and names do not occur.
- **MS7** Every column of every data file is a registered name.
- **MS8** None of the symbols of Section 3.3 occurs in a document of the folder.

6. How to repeat the checks

Run from the folder `Revision_Deflation_CJT_and_Kohn-Sham_definition_C` of the repository:

```
python3 code/scripts/write_symbol_registry.py
python3 code/scripts/check_symbols.py
```

The first command writes `provenance/symbols_master.csv` from three sources: the registry of the flat-spacetime part of the project (`../provenance/symbols.csv`), the registry of the first revision on the deflating spacetimes (`../Revision_Deflation/provenance/symbols_deflation.csv`), and the list of new and reworded entries inside the script. It stops with an error if two entries would have the same symbol or the same program name. The second command performs the eight checks and writes `data/symbol_checks.txt`; its exit status is zero only if all pass.

7. The complete registry

Columns: the symbol as it is written in the documents; the names that programs and data files use for it; its meaning; its origin (the key is in Section 8; "new" marks a symbol introduced by this project, "author" a name fixed by an instruction of the author).

1 Coordinates and indices

symbol	names in programs and data files	meaning	origin
$X = \{x0, \dots, x7\}$	`x0` `x1` `x2` `x3` `x4` `x5` `x6` `x7`	the eight coordinates of both spacetimes, never renamed: $x0$ hidden space-like; $x1, x2, x3$ ordinary space; $x4$ ordinary time; $x5, x6, x7$ additional coordinates (time-like in 4+4, space-like in 7+1)	N01 (text at its head) and N04 `X` (section DEFCHART): the same eight names with the same meaning. N02 uses the same eight names with `x0` as the time: related by swap04
swap_{04}	`swap04`	the exchange of the index values 0 and 4; it carries the labels of `7+1.nb` (time at index 0) into those of the project (time at index 4)	new
A, B, C, D	—	index with the values $0, \dots, 7$; always labels the coordinate xA	new
a, b	—	indices whose range is stated at each place of use: the four extra axes, values 0, 5, 6, 7 (in everything that concerns pairs and the order parameter), the four flat coordinates other than the time, values 0, 1, 2, 3 (in the statements of the machine checks), or the sixteen components of the field, values 1 to 16 (in the density matrix n_{ab}). The letters `a`, `b` in sans-serif type are the scale factors of the Einstein-Lovelock evolution; a_{sp}, a_{sc}, $a(T)$, b_t and b have their own entries	new
i, j	—	index of space-like coordinates; values 1, 2, 3 (ordinary space) unless the range is stated: $j = 0, 1, 2, 3$ for the four flat coordinates other than the time where the deflating coordinates occur as well, and all seven space-like directions in the Hamiltonian of 7+1. In the Einstein-Lovelock equations i, j label the seven space directions	new
h (index)	—	index with the values 1, 2, 3 that labels the 4×4 matrices s_h, t_h	N01, N02, N05 `h` in `s4by4[h]`, `t4by4[h]`

2 Matrices of the 4+4 spacetime (tag 1644)

symbol	names in programs and data files	meaning	origin
η_{1644}	`eta1644`	metric of the flat 4+4 spacetime and matrix of the squares of its Dirac matrices, $\text{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$: space-like $= +1$. It equals the flat metric $\eta^{(4+4)}$ of the author's sign convention for the deflating spacetimes	N01, N02, N05, N10–N12 `η4488`; in N01 also `η8` and `g_AB`; D08 G_{AB} with the index value 8 in the place of the value 0
s_h, t_h	—	the three self-dual and the three anti-self-dual real antisymmetric 4×4 matrices	N01, N02, N05, N10–N12 `s4by4[h]`, `t4by4[h]`; the letters are those of D06 (s^h self-dual, t^h anti-self-dual)
τ^A_{1644}, $\bar{\tau}^A_{1644}$	`tau1644` `taubar1644`	the real 8×8 blocks of the 4+4 Dirac matrices (reduced generators of the split octonion algebra)	N01, N10–N12 `τ[A]` and `τ` with an overbar; N02 `τ[A]`, `τbar[A]`; matrices of this kind are the τ^A, $\bar{\tau}^A$ of D07 and D08, whose index runs from 1 to 8 (the value 8 there is the value 0 here)

symbol	names in programs and data files	meaning	origin
σ_{88}	`sigma88`	the 8×8 matrix with the 4×4 blocks $(0, 1_4;\ 1_4, 0)$	N01 `σ`, `σ88`; N02 `σ`; N05 `σ88`
γ^A_{1644}	`gamma1644`	the eight real 16×16 Dirac matrices of the 4+4 spacetime	N01, N10–N12 `T16^A[A]`; N02 `T16[A]`; the t^A of D08 are built in the same way
Σ^{AB}_{1644}	`Sigma1644`	$\frac{1}{4}[\gamma^A_{1644}, \gamma^B_{1644}]$, the scaled commutators; they span $\mathfrak{so}(4,4)$	new name; the same matrices are `SAB` and `sAB[A][B]` of N10–N12; NOT `sAB` of N02, which is a sign
ω_{1644}	`omega1644`	the product $\gamma^0_{1644}\cdots\gamma^7_{1644}$; real chirality matrix of 4+4, $\mathrm{diag}(-1_8, +1_8)$	N01 `T16^A[8]`; N02 `T16[8]`
P^L_{1644}, P^R_{1644}	`PL1644` `PR1644`	the real chiral projectors $\frac{1}{2}(1 \mp \omega_{1644})$	N01 `P_L`, `P_R`; N02 `PL44`, `PR44`
C^-_{1644}, C^+_{1644}	`Cm1644` `Cp1644`	the products $\gamma^0\gamma^1\gamma^2\gamma^3$ and $\gamma^4\gamma^5\gamma^6\gamma^7$ of the 4+4 matrices; $C^\mp\gamma^A(C^\mp)^{-1} = \mp(\gamma^A)^T$	N01, N02, N10–N12 `σ16` (C^-); N02 `S44` (C^+)
W_{1644}	`W1644`	$i\,C^-_{1644}\gamma^4_{1644}$, the grading: Hermitian, $W^2 = 1$; density of the covariant current	new
K_{1644}	`K1644`	generator of the evolution of dirac1644 in $x4$: $\partial_4\psi = K\psi$	new
β_{1644}, α^j_{1644}	`beta1644` `alpha1644`	the Hermitian matrices $-i\gamma^4_{1644}$ and $-\gamma^4_{1644}\gamma^j_{1644}$ ($j = 0, 1, 2, 3$) of the quantized 4+4 theory	new
h_{1644}	`h1644`	Hamiltonian matrix of the quantized 4+4 theory, iK_{1644} for fields independent of $x5, x6, x7$	new
Π_{1644}	`Pi1644`	the projector $(i\gamma^A_{1644}k_A + m_{1644})/2m_{1644}$ on the plane-wave solutions of the worked example	new

3 Matrices of the 7+1 spacetime (tag 1671)

symbol	names in programs and data files	meaning	origin
η_{1671}	`eta1671`	the matrix of the squares of the Dirac matrices of 7+1, $\gamma^A_{1671}\gamma^B_{1671} + \gamma^B_{1671}\gamma^A_{1671} = 2\,\eta^{AB}_{1671}$: $\mathrm{diag}(-1,-1,-1,-1,+1,-1,-1,-1)$, time at index 4. The flat-space part of the project uses it as the metric (time-like $= +1$). It is minus the flat metric $\eta^{(7+1)}$ of the author's sign convention for the deflating spacetimes (space-like $= +1$)	N02 `η17` with the index values 0 and 4 exchanged; NOT `η71`
τ^A_{1671}	`tau1671`	the real 8×8 matrices of the 7+1 construction: $\tau^4_{1671} = 1_8$, the other seven represent the imaginary octonion units	N02 `τ0[swap04(A)]`
γ^A_{1671}	`gamma1671`	the eight real 16×16 Dirac matrices of the 7+1 spacetime (contravariant index)	N02 `Γ17up[swap04(A)]`; NOT `Γ71[A]`
Σ^{AB}_{1671}	`Sigma1671`	$\frac{1}{4}[\gamma^A_{1671}, \gamma^B_{1671}]$, the scaled commutators; they span $\mathfrak{so}(7,1)$	N02 `Σ17[A,B]`, built from the `Γ17` with lower index: $\Sigma^{AB}_{1671} = \eta^{AA}\eta^{BB}_{1671}$ `Σ17[A', B']` with $A' = \mathrm{swap}_{04}(A)$, $B' = \mathrm{swap}_{04}(B)$
ω_{1671}	`omega1671`	the product $\gamma^0_{1671}\cdots\gamma^7_{1671}$; real, $\omega^2_{1671} = -1$	N02 `Γ17[8]`, `ω17`

symbol	names in programs and data files	meaning	origin
χ_{1671}	`chi1671`	$i\,\omega_{1671}$, the (complex) chirality matrix of 7+1	N02 `χ17`; NOT `χ71`
P^L_{1671}, P^R_{1671}	`PL1671` `PR1671`	the complex chiral projectors $\frac{1}{2}(1 \mp \chi_{1671})$	N02 `PL17`, `PR17`; NOT `PL71`, `PR71`
U^c_{1671}	`Uc1671`	the unitary matrix $(1 - i\gamma^4_{1671})/\sqrt{2}$ that carries the complex projectors of 7+1 into the real projectors of 4+4	new
C^+_{1671}, C^-_{1671}	`Cp1671` `Cm1671`	γ^4_{1671} and $\gamma^4_{1671}\omega_{1671}$; $C^\pm\gamma^A(C^\pm)^{-1} = \pm(\gamma^A)^T$	new
β_{1671}, α^j_{1671}	`beta1671` `alpha1671`	the real symmetric matrices γ^4_{1671} and $\gamma^4_{1671}\gamma^j_{1671}$ (seven space indices j) of the Hamiltonian	N02 `βD`, `αD[swap04(j)]`
h_{1671}	`h1671`	Hamiltonian matrix of the 7+1 theory, $\alpha^j_{1671}p^j + \beta_{1671}m_{1671}$	N02 `h`
Π_{1671}	`Pi1671`	the projector $(\gamma^A_{1671}k_A + m_{1671})/2m_{1671}$ of the worked example	new
E^L_{1671}, E^R_{1671}	`EL1671` `ER1671`	the chiral frames: 16×8 matrices whose columns span the two eigenspaces of χ_{1671}	N02 `EL`, `ER`
M_{1671}, U^+_{1671}, V^-_{1671}	`M1671` `Upos1671` `Vneg1671`	the 8×8 matrix $m_{1671} + p^j\tau^j_{1671}$ and the matrices of positive- and negative-energy spinors of the worked example (check `71-D8`)	N02 `M` of its Sec. 7.4 (in N02 the letter `M` also denotes other matrices and a mass); the two spinor matrices are new names

4 Fields and their equations

symbol	names in programs and data files	meaning	origin
ψ_{1644}	`dirac1644`	the field dirac1644: 16 complex Grassmann components on the 4+4 spacetime; $(\gamma^A_{1644}\partial_A - m_{1644})\psi_{1644} = 0$	new; `Ψ16` of N01 and N10–N12 is a column of 16 real quantities (in N10–N12 functions of `x0`, `x4` called the wave function of the universe), not this field
ψ^L_{1644}, ψ^R_{1644}	`psiL1644` `psiR1644`	the two 8-component chiral parts of dirac1644 (upper and lower eight components)	new
m_{1644}	`m1644`	mass of the field dirac1644	new
$\tilde{\psi}$	—	the covariant adjoint $\psi^\dagger C^-_{1644}$ of the 4+4 theory (invariant under $\mathrm{Spin}_0(4,4)$)	new
j^A_{1644}	`j1644`	the conserved current $i\,\tilde{\psi}\gamma^A_{1644}\psi$ of the covariant 4+4 theory (fermion number)	new
ψ_{1671}	`dirac1671`	the field dirac1671: 16 complex Grassmann components on the 7+1 spacetime; $(i\gamma^A_{1671}\partial_A - m_{1671})\psi_{1671} = 0$	N02 `dirac16complex`
ϕ^L_{1671}, ϕ^R_{1671}	`phiL1671` `phiR1671`	the two 8-component chiral parts of dirac1671	N02 `φL`, `φR`
m_{1671}	`m1671`	mass of the field dirac1671	N02 `m`, `mD`
$\bar{\psi}$	—	the adjoint $\psi^\dagger\beta$ of the quantized theories ($\beta = \beta_{1671}$ or β_{1644}); in 4+4 it differs from $\tilde{\psi}$	new
β, α^j, χ, h, m	—	in the sections common to both spacetimes: the matrices and the mass of the spacetime at hand, i.e. the tagged symbols with the tag dropped; $\chi \equiv \chi_{1671}$ in 7+1 and $\chi \equiv \omega_{1644}$ in 4+4	new

symbol	names in programs and data files	meaning	origin
$U(\vec{p})$	—	the 16×8 matrix of orthonormal positive-energy spinors at momentum $\vec{p}$	new

5 Symmetries, pairs and order parameter

symbol	names in programs and data files	meaning	origin
$U(1)_F$	—	the phase symmetry $\psi \to e^{i\epsilon}\psi$: fermion number	new
$U(1)_{VS}$	`UVS`	the chiral symmetry $\psi \to e^{i\epsilon\chi}\psi$, $\Phi_{VS} \to e^{-2i\epsilon}\Phi_{VS}$, broken by the vacuum value of the Vorton-Scalar	new
$SU(2)_A$, $SU(2)_B$	—	the two factors of the internal symmetry of the fields that do not depend on the four extra coordinates (zero-mode sector), in both spacetimes; $SO(4) = [SU(2)_A \times SU(2)_B]/Z_2$	new
$SU(2)_X$, $SU(2)_Y$	—	the two factors of the rotation group of the four space coordinates $x0, \dots, x3$ of the 4+4 spacetime (NOT the internal $SU(2)_A \times SU(2)_B$)	new
$SO(3)_T$, $SU(2)_T$	—	the rotations of the three additional times $x5, x6, x7$ of 4+4, and its covering group; the diagonal subgroup of $SU(2)_A \times SU(2)_B$	new
$U(1)_E$, P (generator)	—	the second central phase symmetry of the zero-mode sector and its traceless generator, $P^2 = 1$: $P = +1$ on the doublets of $SU(2)_A$, -1 on those of $SU(2)_B$	new
Δ^A_{1644}, Δ^A_{1671}	`DeltaV1644` `DeltaV1671`	the vector pair operators $\psi^T C^- \gamma^A \psi$ of each spacetime	new
Δ^{AB}_{1671}	`DeltaT1671`	the two-form pair operators $\psi^T C^+ \gamma^A \gamma^B \psi$ of the 7+1 spacetime	new
Δ^a	—	order parameter: expectation value of the vector pair operator along the extra axis $a \in \{0, 5, 6, 7\}$; $\lvert\Delta\rvert^2 \equiv \sum_a \lvert\Delta^a\rvert^2$	new
$\Phi_X(\vec{p})$	—	two-particle content $U(\vec{p})^T X U(-\vec{p})$ of the pair operator $\psi^T X \psi$	new
`M`	`M_gap`	the 8×8 gap matrix on the internal states of a pair (sans-serif; `M_gap` in the statements of the programs)	new
$d_\pm$	—	the two fourfold gaps of a general state, $d^2_\pm = \lvert\Delta\rvert^2 \pm \sqrt{\lvert\Delta\rvert^4 - \lvert\eta_{ab}\Delta^a\Delta^b\rvert^2}$	new
n^a	—	real unit vector: direction of the order parameter in the space of the four extra axes	new

6 The Vorton-Scalar (suffix VS)

symbol	names in programs and data files	meaning	origin
Φ_{VS}	`PhiVS`	the Vorton-Scalar: a complex scalar field	new (the name is that of the author's project NCSS, where the scalar is real)
v_{VS}	`vVS`	vacuum value: $\langle\Phi_{VS}\rangle = v_{VS}/\sqrt{2}$	new
λ_{VS}	`lambdaVS` `lamVS`	self-coupling of the Vorton-Scalar	new
y_{VS}	`yVS`	Yukawa coupling of the Vorton-Scalar to the fermion field	new

symbol	names in programs and data files	meaning	origin
s_{VS}	`sVS`	the radial mode of the Vorton-Scalar (real field)	new
m_s	`msVS`	mass of the radial mode, $\sqrt{2\lambda_{VS}}\,v_{VS}$	new
ϑ_{VS}	`thetaVS thVS`	the phase mode of the Vorton-Scalar (real field, massless)	new
g_{VS}	`gVS`	coupling of both modes of the Vorton-Scalar to the fermions, $g_{VS}=y_{VS}/\sqrt{2}=m_Y/v_{VS}$, where m_Y is the part of the fermion mass that the Vorton-Scalar gives (equal to m/v_{VS} when there is no bare Dirac mass)	new
α_{VS}	`alphaVS aVS`	$g_{VS}^2/4\pi$	new
D_{VS}	`DVS`	$\alpha_{VS}\,m/m_s$: two fermions bind if and only if $D_{VS}>D_c$	new
J^A_{VS}	`JVS`	the neutral current of the Vorton-Scalar (bilinear in Φ_{VS})	new
J^A_5	`J5`	the axial current $\bar\psi\gamma^A\chi\psi$ of the fermion field	new
G_{VS}	`GVS`	constant of the Fermi type of the neutral-current interaction, $G_{VS}/\sqrt{2}=1/(2v_{VS}^2)$	new
$\mathcal{L}^{VS}_{1644}$, $\mathcal{L}^{VS}_{1671}$	`LVS`	the Lagrangians that define the Vorton-Scalar in the two spacetimes	new
C (coupling)	—	$g_{VS}^2m^2/m_s^2=g_{VS}^4/2\lambda_{VS}$, the one number on which the fermion mass in matter depends	new

7 Pairing, gap equation and CJT effective action

symbol	names in programs and data files	meaning	origin
ν	—	the number of internal states of a particle of given momentum, $\nu=8$	new
F_S, F_P	`F_scalar F_pseudoscalar`	exchange amplitudes of the radial and of the phase mode in a pair channel	new
$V_{\rm NC}$, $V_s(r)$	—	contact interaction of the neutral current in the hidden-axis channel; Yukawa potential of the radial mode	new
σ_X	—	the sign $+1$ (vector pairs along the extra axes) or -1 (their two-form pairs) of the phase-mode term of the kernel	new
$W(k,k')$	—	kernel of the gap equation: the s-wave average of the pairing attraction, $W(k,k')=\frac{1}{2}\int_{-1}^{1}d\cos\theta\,[-V_{VS}-V_{\rm ct}]$ for the momenta k, k' of the pair before and after the scattering (θ: the angle between them); without contact couplings the exchange attraction alone. Always written with its two arguments (W_{1644} is a matrix)	new
$\Delta(k)$, Δ_F	—	gap function; its value at the Fermi momentum	new
ξ_k, E_k, $\kappa(k)$	—	$\sqrt{m^2+k^2}-\mu$; quasi-particle energy $\sqrt{\xi_k^2+\Delta(k)^2}$; pair correlation $\langle b(-\vec k)b(\vec k)\rangle$	new
k_F, E_F, μ	`kF mu`	Fermi momentum; Fermi kinetic energy $\sqrt{m^2+k_F^2}-m$; chemical potential. The number density is $n=\nu V_D k_F^D/(2\pi)^D$ in D space dimensions ($n=\nu k_F^3/6\pi^2$ for $D=3$)	new
$N(0)$	—	density of states at the Fermi surface per internal state, $k_F\sqrt{m^2+k_F^2}/2\pi^2$	new
λ, λ_S, λ_P	—	dimensionless pairing coupling $N(0)W(k_F,k_F)$ and its parts from the radial and the phase mode	new
T_c	—	critical temperature of the paired state	new

symbol	names in programs and data files	meaning	origin
D_c	—	critical value of D_{VS} for a bound pair in the Yukawa potential	new
a_{sc}, a_{BB}	—	scattering length of two fermions; scattering length of two bound pairs	new
$\Gamma[\phi, G]$, Γ_2	—	the effective action for composite operators of Cornwall, Jackiw and Tomboulis; its two-particle-irreducible part	new (method of the cited paper)
$\mathcal{S}$, D_s, D_ϑ	—	propagator of the doubled fermion field; propagators of the radial and of the phase mode	new
$V[\Delta]$	—	the CJT effective potential of the paired state, measured from the unpaired state	new
$\eta(T)$	—	largest eigenvalue of the linearized gap equation, $\eta(T_c) = 1$	new
$A_2(T)$, A_4	—	coefficients of the expansion of the CJT effective potential V along a gap function of fixed shape	new
$a(T)$, b, K_{GL}	—	Ginzburg-Landau coefficients of the weak-coupling free energy (quadratic, quartic and gradient term)	new
c_s	—	speed of first sound of the superfluid, $c_s^2 = dP/d\rho$	new

8 Matter, temperature and vacuum

symbol	names in programs and data files	meaning	origin
ϕ	—	expectation value of the Vorton-Scalar, $\langle \Phi_{VS} \rangle = \phi/\sqrt{2}$; the fermion mass is $g_{VS}\phi$	new
m^*, x	`M` `M_ks` `M_cjt` `M_cjt_global` `M_plus` `M_minus` `M_relaxed` `x` `Mp_ks` `Mp_cjt` `Mm_ks` `Mm_cjt` `Mp_below` `Mm_below` `Mp_above` `Mm_above` `M_large` `M_small`	m^*: the mass of the fermions in matter; $x = \phi/v_{VS}$: the expectation value of the Vorton-Scalar in units of its vacuum value. With a contact interaction $m^* = m + G_S\, n_S$ (gap equation of the Kohn-Sham and CJT calculations); with the Vorton-Scalar and a bare Dirac mass $m^* = m_D + m_Y\, x + G_S\, n_S$, so that $x = m^*/m$ if the whole mass comes from the Vorton-Scalar and no contact coupling acts. m^*_+, m^*_-: the masses of the particles of the two halves $W_{1644} = \pm 1$ of the 4+4 field, $m^*_\pm = m + G_S^{\mathrm{tot}}\, n_S \pm G_S^{\mathrm{dif}}\, n_{SW}$ in both forms of the self-interaction (with the filled states of negative energy: the masses of the two halves in vacuum). The data files and the statements of the programs write m^* as `M` and $m^*_\pm$ as `M_plus`, `M_minus` (`Mp_..`, `Mm_..` with a suffix that names the method or the strength; `M_large`, `M_small`: the larger and the smaller of the two); the author's document KOHN_SHAM_GRAVITY writes M	new
$\varepsilon(k_F, x)$, $V_{\mathrm{sea}}(x)$	—	energy density of matter as a function of the Fermi momentum and of $x = m^*/m$ (in units of m^4 in ordinary space); renormalized energy of the filled negative-energy states. ε without arguments is the energy density of matter in the units of the calculation at hand	new
B	—	energy density of the state without expectation value of the Vorton-Scalar (bag constant)	new
E_x	—	exchange (Fock) energy of the two modes	new
M_s, M_ϑ, a_s, a_ϑ	—	thermal masses of the two boson modes in the Hartree approximation, and the curvatures U''_{eff}, U'_{eff}/ϕ	new
$P(M)$, $Q(M)$	—	thermal part of the boson loop and of $\frac{1}{2}\mathrm{Tr}\ln$	new
T_{high}, T_N, $T_{1/2}$, T_0	—	two closed estimates of the temperature at which the fermion mass vanishes; temperatures at which the expectation value has fallen to half and to zero	new

symbol	names in programs and data files	meaning	origin
$\sigma(p)$	—	self-energy that a pair condensate in the vacuum gives to the fermions: mass splitting $m \pm \sigma$	new
α_c, Λ (cutoff)	`Lambda`	critical value of α_{VS} for a pair condensate in the vacuum; Λ: a momentum cutoff (of the ladder equation in the vacuum, and of the filled states of negative energy in the calculations with contact interaction). Λ_8 and $\bar{\Lambda}$ are cosmological terms and have their own entries	new

9 Cosmology

symbol	names in programs and data files	meaning	origin
N	—	density of the dark radiation today in units of the density of one massless neutrino species	new
z_t, T_d	—	transition redshift of the stepped dark fluid, $m = T_{d0}(1 + z_t)$; temperature of the dark sector	new
r_{step}	—	factor by which the density of the dark radiation rises when the fermions annihilate, $15^{1/3}$ for this model	new
ω_b, ω_c, h, H_0, A_s, n_s, τ	`tau_reio`	the six parameters of the standard cosmological model: densities of baryons and of cold dark matter (Ωh^2), Hubble constant $H_0 = 100\,h$ km s^{-1} Mpc^{-1}, amplitude and spectral index of the primordial fluctuations, optical depth of reionization (`tau_reio` in the data files; the letter τ with the index d is the deflation clock)	standard notation
$\Delta\chi^2$	—	worsening of the best fit to the mock data relative to the standard model	new
θ_s, r_s, D_A	—	angular size of the sound horizon; sound horizon; distance to last scattering	standard notation
c_{s0}	—	sound speed of superfluid dark matter today at the mean cosmic density	new
ρ_0, R_{TF}	—	constant of the equation of state of a condensate of bound pairs; Thomas-Fermi radius of a core supported by its pressure	new
m_{deg}, $m_{B,\text{max}}$	—	largest fermion mass for degeneracy, $m_{\text{deg}} = [6\pi^2\hbar^3\rho/(\nu\sigma_v^3)]^{1/4}$, and largest mass m_B of a boson for condensation, $m_{B,\text{max}} = [(2\pi)^{3/2}\hbar^3\rho/(2.612\,\sigma_v^3)]^{1/4}$, in a halo of the mass density ρ and the velocity dispersion σ_v	new
σ_v; κ_{SI}	—	the velocity dispersion of the dark matter of a halo (with the index v; σ_8 is the amplitude of clustering and $\sigma(p)$ a self-energy); the bound on the cross-section per mass of dark matter from colliding clusters, about 1 cm^2/g (with the index SI; κ_8 is the gravitational coupling)	new

D1 Deflating spacetimes: hypotheses and geometry

symbol	names in programs and data files	meaning	origin
g55, g66, g77	`g55` `g66` `g77`	the three rates of deflation of $x5$, $x6$, $x7$ (positive constants, dimension 1/length). These are the author's names; they are written upright and never with a subscript. They are *not* components of the metric: the components are g_{55}, g_{66}, g_{77}	author
rate_t	`rate_t` `rate`	the rate of the coordinate xt in a statement that holds for $t = 5, 6, 7$ alike: $\mathrm{rate}_5 = \mathrm{g55}$, $\mathrm{rate}_6 = \mathrm{g66}$, $\mathrm{rate}_7 = \mathrm{g77}$	new
gsum	`gsum` `gam`	$\mathrm{g55} + \mathrm{g66} + \mathrm{g77}$, the total rate: the proper volume of the three deflating coordinates falls as $e^{-\mathrm{gsum}\, x4}$ (`gam`: gsum in units of 100 km/s/Mpc)	new
gsq, gprod	`gsq` `gprod`	$\mathrm{g55}^2 + \mathrm{g66}^2 + \mathrm{g77}^2$ and $\mathrm{g55}\,\mathrm{g66} + \mathrm{g55}\,\mathrm{g77} + \mathrm{g66}\,\mathrm{g77}$; $\mathrm{gsum}^2 = \mathrm{gsq} + 2\,\mathrm{gprod}$	new
const5, const6, const7; const_t	`const5` `const6` `const7` `const_t` `cst`	the three positive constants of the author's metric (his word: const): the values of the scale factors at $x4 = 0$	author
b_5, b_6, b_7; b_t	`b_5` `b_6` `b_7` `b_t`	the scale factors of the deflating coordinates, $b_t(x4) = \mathrm{const}_t\, e^{-\mathrm{rate}_t\, x4}$: a coordinate distance dxt is the proper length $b_t\,\lvert dxt\rvert$. Always written with the index 5, 6, 7 or t (the letter b without index is a Ginzburg-Landau coefficient, and b in sans-serif type is the common scale factor of the three deflating coordinates in the Einstein-Lovelock evolution)	new
V_d	`Vd`	$b_5 b_6 b_7 = \sqrt{\lvert\det g\rvert}$, the proper volume of a unit coordinate cell of the deflating coordinates; $V_d \propto e^{-\mathrm{gsum}\, x4}$	new
μ, ν, ρ, σ (indices)	—	coordinate indices with the values $0, \ldots, 7$ on the deflating spacetimes (only as sub- or superscripts; as stand-alone letters μ is the chemical potential, ν the number of internal states and λ the pairing coupling). The capital indices A, B, C, D, also with a prime, label the orthonormal frame: the frame vector A points along the coordinate xA (as a stand-alone letter C is the coupling $g_{VS}^2 m^2/m_s^2$, and D the number of space dimensions with momentum). In the statements of the programs the indices are spelled out inside tensor expressions, as in `g^(mu nu)`	new
t, s (indices)	—	indices of the three additional coordinates $x5, x6, x7$, values 5, 6, 7; s is a second such index where two are needed (the matrices s_h carry the index h)	new
${e_A}^{\mu}$, ${e^A}_{\mu}$	`e_A^mu`	the orthonormal frame and its inverse: ${e^A}_{\mu} = \mathrm{diag}(1, 1, 1, 1, 1, b_5, b_6, b_7)$	new
${\Gamma^{\rho}}_{\mu\nu}$	`Gamma` `christoffel`	the Christoffel symbols of the deflating metric (three indices; $\Gamma[\phi, G]$ with square brackets is the CJT effective action). The author's document KOHN_SHAM_GRAVITY writes the Christoffel symbols with braces and uses Γ^A for the Dirac matrices, which are γ^A here	new
${R^{\rho}}_{\sigma\mu\nu}$, ${R^{\mu}}_{\nu}$, R	`riemann` `ricci` `Ric44` `Ric55` `Ric66` `Ric77` `Rscalar`	Riemann tensor, Ricci tensor and curvature scalar of the deflating metric	new
${\mathcal{G}^{\mu}}_{\nu}$	`Ein00` `Ein44` `Ein55` `Ein66` `Ein77` `einstein_mixed`	the Einstein tensor ${R^{\mu}}_{\nu} - \frac{1}{2} R\, {\delta^{\mu}}_{\nu}$ (script letter, so that it cannot be taken for Newton's constant or for a constant of the Fermi type)	new
$\mathcal{X}$; $\mathcal{X}_{\mathrm{scale}}$, $\mathcal{X}_{(s)}$	`killing_vectors` `killing_count`	a Killing vector field of the deflating metric (script letter; the plain letter X is the set of the eight coordinates); the scaling vector $\partial_4 + \sum_t \mathrm{rate}_t\, xt\, \partial_t$; the three special Killing vectors of the case of equal rates, $s = 5, 6, 7$	new
$\bar{x}4$; $\bar{x}5$, $\bar{x}6$, $\bar{x}7$	—	coordinates of the case of equal rates in which the line element of $x4, \ldots, x7$ is a multiple of a flat one: $\bar{x}4 = e^{\mathrm{g55}\, x4}/\mathrm{g55}$, $\bar{x}t = \mathrm{const}_t\, xt$	new

symbol	names in programs and data files	meaning	origin
$\mathcal{H}^{\mu}{}_{\nu}$; α_{GB}, Λ_8	`Lanczos` `alpha_GB` `Lambda_8`	the Lanczos tensor of the Gauss-Bonnet term, $\mathcal{H}^{\mu}{}_{\nu} = 2(R\,R^{\mu}{}_{\nu} - 2R^{\mu}{}_{\alpha}R^{\alpha}{}_{\nu} - 2R^{\mu\alpha}{}_{\nu\beta}R^{\beta}{}_{\alpha} + R^{\mu\alpha}{}_{\beta\gamma}R^{\beta\gamma}{}_{\nu\alpha}) - \frac{1}{2}\delta^{\mu}_{\nu}(R^2 - 4R^{\alpha}{}_{\beta}R^{\beta}{}_{\alpha} + R^{\alpha\beta}{}_{\gamma\delta}R^{\gamma\delta}{}_{\alpha\beta})$ (script letter; H is the Hubble rate); the coupling of the Gauss-Bonnet term and the cosmological constant of Einstein-Gauss-Bonnet gravity in eight dimensions, whose equations without matter are $\mathcal{G}^{\mu}{}_{\nu} + \Lambda_8\,\delta^{\mu}_{\nu} + \alpha_{GB}\,\mathcal{H}^{\mu}{}_{\nu} = 0$ (metric with positive space-like components). In the author's Einstein-Lovelock notation $E^{(1)} = -\mathcal{G}$, $E^{(2)} = \mathcal{H}$, $\alpha_2 = -\alpha_{GB}$, $\bar{\Lambda} = \Lambda_8$	new
K_{AB}	`ksec`	the sectional curvature of the plane of the frame directions A and B (metric with positive space-like components): on the deflating spacetimes $K_{4t} = \text{rate}_t^2$, $K_{ts} = \text{rate}_t\,\text{rate}_s$ for $t \neq s$, zero if one direction is flat; for scale factors that depend on $x4$ in any way, $K_{4i} = H_i' + H_i^2$ and $K_{ij} = H_iH_j$ with the rate H_i of the direction i (the pair curvatures of the author's Einstein-Lovelock equations). Always with two subscripts: the letter K without subscript is the generator of the evolution of the 4+4 field in $x4$, K as an index is an index set of the self-interaction, and K with an order as subscript and with an argument is a modified Bessel function	new
ς	`affine`	affine parameter along a geodesic; for a massive particle on the 7+1 spacetime, the proper time divided by the mass	new
P_{μ}	`P_1` `P_5`	the conserved covariant momenta of a geodesic (with an index; P without index is the generator of $U(1)_E$)	new
$x4_{\text{turn}}$	`x4_turn`	the value of the ordinary time at which a classical particle with momentum along a deflating time turns back, and beyond which a mode that depends on a deflating time is extinguished (4+4): there $k_d = E_{\perp}$	new

D2 Deflating spacetimes: matrices, fields and their equations

symbol	names in programs and data files	meaning	origin
γ^{μ}_{1644d}, γ^{μ}_{1671d}	`gamma_1644d` `gamma_1671d`	the Dirac matrices of the deflating spacetimes in coordinate components, $\gamma^{\mu} = e_A{}^{\mu}\gamma^A$: $\gamma^t = \gamma^{A=t}/b_t$ for $t = 5, 6, 7$, the other five are the constant matrices γ^A_{1644}, γ^A_{1671} of the frame	new
$\omega_{\mu AB}$	`omega_(mu A B)`	the coefficients of the spin connection, $\omega_{\mu AB} = \eta_{AC}\,e^C{}_{\nu}(\partial_{\mu}e_B{}^{\nu} + \Gamma^{\nu}{}_{\mu\lambda}e_B{}^{\lambda})$ with the flat metric $\eta = \eta^{(4+4)}$ or $\eta^{(7+1)}$ (three indices; ω_{1644}, ω_{1671} with a tag are the products of all eight Dirac matrices)	new
Ω_{μ}	`Omega_mu` `omega`	the spin connection as a matrix, $\Omega_{\mu} = \frac{1}{4}\,c_{\gamma}\,\omega_{\mu AB}\gamma^A\gamma^B$; the covariant derivative of a spinor is $\nabla_{\mu}\psi = (\partial_{\mu} + \Omega_{\mu})\psi$. On both deflating spacetimes $\Omega_t = \frac{1}{2}\,\text{rate}_t\,b_t\,\gamma^t\gamma^4$ (frame matrices) for $t = 5, 6, 7$ and $\Omega_{\mu} = 0$ otherwise. (Ω_{CJT} is an effective potential)	new
ψ_{1644d}, ψ_{1671d}	`dirac1644d` `dirac1671d`	the fields dirac1644d and dirac1671d: the 16-component fields on the deflating spacetimes. With mass term and self-interaction they obey $(\gamma^{\mu}\nabla_{\mu} - m)\psi_{1644d} = \sum_r g_r \sum_{\lvert K\rvert=r}(\tilde{\psi}\Theta_K\psi)\Theta^K\psi$ (covariant form of 4+4) and $(i\gamma^{\mu}\nabla_{\mu} - m)\psi_{1671d} = \sum_r g_r \sum_{\lvert K\rvert=r}(\bar{\psi}\Theta_K\psi)\Theta^K\psi$ (7+1)	new

symbol	names in programs and data files	meaning	origin
$\hat{\psi}$	psihat	the weighted field $\sqrt{V_d}\,\psi$, which obeys the equation without spin connection	new
$K(x4)$, $\mathsf{h}(x4)$	K h	the generator K_{1644} and the Hamiltonian matrix h_{1671} with the wavenumbers along the deflating coordinates divided by the scale factors: without self-interaction $\partial_4\hat{\psi} = K\hat{\psi}$ (4+4), $i\partial_4\hat{\psi} = \mathsf{h}\hat{\psi}$ (7+1)	new

D3 Deflating spacetimes: solutions mode by mode

symbol	names in programs and data files	meaning	origin
k_5, k_6, k_7; k_t	k_5 k_6 k_7 k_t	the conserved wavenumbers of a mode along the deflating coordinates (covariant components of the wave vector k_A)	new
$E_\perp$	E_perp	the energy of a mode, or of a classical particle, without its motion along the deflating coordinates: $\sqrt{m^2 + k_0^2 + k_1^2 + k_2^2 + k_3^2}$ (with the wavenumbers or momenta along $x0, \dots, x3$)	new
k_d	k_d	the physical wavenumber along the deflating coordinates, $k_d(x4) = \left[\sum_t k_t^2/b_t(x4)^2\right]^{1/2}$; it grows as the coordinates deflate	new
$\mathsf{h}_\perp$, $K_\perp$	h_perp K_perp	the parts of h and of K that do not contain the scale factors: the operators of the modes that do not depend on the deflating coordinates, $\mathsf{h}_\perp = m\beta + \sum_a p_a\alpha^a$ with $\mathsf{h}_\perp^2 = E_\perp^2$, and $K_\perp = -m\gamma^4 + i\sum_a k_a\gamma^4\gamma^a$ with $K_\perp^2 = -E_\perp^2$	new
γ^d, α^d	—	the frame matrix along the direction of the wave vector in the deflating coordinates, $\gamma^d = k_d^{-1}\sum_t (k_t/b_t)\gamma^t$, and $\alpha^d = \gamma^4\gamma^d$; constant in time when one rate governs	new
w; f_+, f_-; Z_+, Z_-	—	auxiliary objects of the exact solutions of the equations without self-interaction: a constant eigenvector of $\mathsf{h}_\perp$ or of $K_\perp$; the two coefficient functions of the ansatz; two cylinder functions whose orders differ by one	new
u_d	u_d	k_d/rate_t for a mode that depends on one deflating coordinate: the argument of the Bessel functions of the exact solutions	new
n_{cr}	n_cr n_min n_mean n_max n_exact fd5 fd6	the number of particles per internal state that the deflation creates in a mode that starts empty (7+1)	new
T_{defl}	—	the temperature of the thermal spectrum of the created quanta, $\mathrm{rate}_t/2\pi$ in units with $\hbar = k_B = 1$ (7+1)	new
S_{WKB}	S_wkb	$\int \sqrt{k_d^2 - E_\perp^2}\, dx4$ from the turning value on: exponent of the extinction of a bounded solution (4+4)	new
φ_{bos}, $\hat{\varphi}_{\mathrm{bos}}$, m_{bos}	phi_bos Shat Shat2 m_bos	a mode of the Vorton-Scalar (the radial mode, $m_{\mathrm{bos}} = m_s$, or the phase mode, $m_{\mathrm{bos}} = 0$), its weighted form $\sqrt{V_d}\,\varphi_{\mathrm{bos}}$, and its mass	new
ϖ_b	varpi_b	$\sqrt{k^2 + m_{\mathrm{bos}}^2 - \mathrm{gsum}^2/4}$: the frequency of the weighted boson mode that does not depend on the deflating coordinates	new

D4 The deflation clock and the couplings

symbol	names in programs and data files	meaning	origin
τ_d	`tau_d`	the deflation clock, $\tau_d = \mathrm{gsum}\,(x4 - x4_{\mathrm{ref}})$: the number of e-foldings by which the volume V_d has shrunk since the reference time; in an evolution in which the rates change, $\tau_d = -\ln(V_d/V_d^{\mathrm{ref}})$ (the letter τ without the index d is the optical depth of reionization)	new
τ_{pair}	`tau_pair`	the clock measured from the moment at which two fermions become bound: $\ln(D_{VS}/D_c)$	new
$\hat{\Phi}_{VS}$	`Phihat`	the weighted Vorton-Scalar $\sqrt{V_d}\,\Phi_{VS}$	new

D5 Dynamics of the paired state

symbol	names in programs and data files	meaning	origin
Ξ_d	`Xi_d`	$\mathrm{gsum}/(\lambda\,\Delta_{\mathrm{eq}})$: decides whether the paired state can follow the deflation ($\Xi_d \ll 1$: yes)	new
n_k	—	occupation of the pair of states of the single-particle energy ξ_k in the shell model of the paired state (with the index k; the letter n without index is a density)	new
Δ_{eq}	`gap_eq`	the equilibrium gap for the pairing strength of the moment	new
E_{sh}	`E_sh`	half-width in energy of the shell of states around the Fermi surface in the model of the pair dynamics; unit of energy and of inverse time there	new
n_{ex}	`n_ex` `n_ex_final`	density of broken pairs (in units of E_{sh} times the density of states)	new

D6 Cosmology and bounds on the deflating spacetimes

symbol	names in programs and data files	meaning	origin
G_N, G_8	`G_N` `GN_early` `GN_star` `G_8`	Newton's constant of ordinary space, and the gravitational coupling of the eight-dimensional Einstein theory (assumption A2): $G_N = G_8/(\text{proper volume of the extra coordinates}) \propto e^{\mathrm{gsum}\,x4}$	new
a_{sp}, H	—	scale factor of ordinary space (assumption A1) and its rate $H = (da_{\mathrm{sp}}/dx4)/a_{\mathrm{sp}}$; H_0 is its value today. (The rates of the Einstein-Lovelock evolution are H_{a}, H_{b}; the author's Einstein-Lovelock notebooks use the letter H for the rate in the exponent of their metric, which corresponds here to g55, g66, g77)	new
t_0	`age` `age_Gyr`	the age of the universe in the model at hand	new
θ_*, z_*	`theta_star` `z_star`	the angle under which the sound horizon at last scattering is seen, $r_s(z_*)/D_A(z_*)$ with the comoving distance D_A, and the redshift z_* of last scattering at which the optical depth without reionization is 1 (the $100\,\theta_*$ of the Planck papers). The angle θ_s is the same ratio at the maximum of the visibility function; for the model without deflation $100\,\theta_* = 1.03965$ and $100\,\theta_s = 1.040358$	new
δ_m, f_g, σ_8	`growth` `f_growth` `growth_ratio` `f_growth_ratio` `f_sigma8_ratio`	density contrast of matter in ordinary space; its growth rate $d\ln\delta_m/d\ln a_{\mathrm{sp}}$ today; the amplitude of clustering (only ratios of σ_8 are used)	new

symbol	names in programs and data files	meaning	origin
v_{EW}, n_{EW}	`v_EW` `n_EW`	the electroweak scale (the vacuum value of the Higgs field of the Standard Model, 246 GeV), and the exponent with which it would follow the coupling under the hypothesis H2: $n_{\mathrm{EW}} = d\ln v_{\mathrm{EW}}/d\ln\alpha_{VS}$	new
ϱ_c	`rho_c`	abbreviation in the formula for n_{EW} of the scale-invariant ladder equation: $\varrho_c = \frac{1}{2}\sqrt{\alpha_{VS}/\alpha_c - 1}$ (the letter ρ without index is a density)	new
m_p/m_e	`mu_pe`	the ratio of the proton mass to the electron mass (always written as this ratio in the documents; `mu_pe` in the statements of the programs)	new
G_{Fermi}, α_{em}	`G_Fermi`	the Fermi constant of the weak interaction and the fine-structure constant (they enter only in the discussion of bounds)	new

D7 Symmetries and vortex lines on the deflating spacetimes

symbol	names in programs and data files	meaning	origin
P_c	`Pc`	$i\alpha^1\alpha^2\alpha^3\chi$, the product of the five matrices of the sector that does not depend on the extra coordinates; equal, up to sign, to the generator P of $U(1)_E$	new
θ_{12}	—	the angle around a vortex line that lies along $x3$, in the plane of $x1$ and $x2$ (with the index 12; θ_* is the angle of the sound horizon and $\theta(\cdot)$ the step function)	new
w_z	`w_z`	a constant spinor of a fermion zero mode on a vortex line	new

D8 Tags and identifiers of the deflating spacetimes

symbol	names in programs and data files	meaning	origin
1644d, 1671d	`1644d` `1671d`	tags of the deflating 4+4 and 7+1 spacetimes and of every object defined on them (the tags 1644, 1671 of the flat spacetimes with the letter `d`)	new
A1, A2, A3, H2, HV	—	A1, A2: the two assumptions of the cosmological part (ordinary space expands; gravity is the eight-dimensional Einstein theory). A3: the assumption about the perturbations of ordinary space (the scale factors of the extra coordinates are not perturbed, and the agent that holds them has no energy density), with its closures. H2: the author's hypothesis that the superfluid state is the Higgs field. HV: the different assumption that the vacuum value of the Vorton-Scalar sets the electroweak scale	new
"Momentum", "Pressure", "Shear" (closures)	`Momentum` `Pressure` `Shear` `closure`	the three closures of the perturbation equations under assumption A3, named after the Einstein equation of ordinary space that holds with matter and radiation alone besides the energy constraint: the momentum constraint, the trace (pressure) equation, or the equation of the shear. In data files the suffixes `_M` and `_P` mark the first two; `_N4` marks the four-dimensional growth equation, which is none of them	new

D9 Perturbations of ordinary space and the microwave background

symbol	names in programs and data files	meaning	origin
$\tau_{\rm cf}$	`tau_cf` `tau0_Mpc`	the conformal time of ordinary space, $d\tau_{\rm cf} = dx4/a_{\rm sp}$, in Mpc (`tau0_Mpc`: its value today). In the equations of the perturbations a prime is the derivative with respect to $\tau_{\rm cf}$	new
$H_{\rm cf}$, H_e	`Hc` `He` `hub` `hub_e` `a_gsum_over_Hc`	the conformal Hubble rate $H_{\rm cf} = a_{\rm sp}H$, and $H_e = H_{\rm cf} - a_{\rm sp}\,$gsum$/2$, which takes its place in three of the Einstein equations of ordinary space when the extra coordinates deflate	new
k (perturbations), ℓ, D_ℓ^{TT}, D_ℓ^{TE}, D_ℓ^{EE}	—	the comoving wavenumber of a perturbation of ordinary space (in Mpc^{-1}; the physical wavenumber is $k/a_{\rm sp}$); the multipole; and the spectra of the temperature, of the polarization and their cross spectrum, $D_\ell = \ell(\ell+1)C_\ell/2\pi$ in μK^2	new
$h_{\rm sy}$, $\eta_{\rm sy}$, $\alpha_{\rm sy}$	—	the two functions that describe a scalar perturbation of the metric of ordinary space in the synchronous gauge (the h and η of Ma and Bertschinger; here with the index sy, because h is the Hubble parameter and η the flat metric), and the combination $\alpha_{\rm sy} = (h'_{\rm sy} + 6\eta'_{\rm sy})/2k^2$	new
$\mathcal{U}$	`potential`	$\eta_{\rm sy} + \alpha'_{\rm sy}$, the sum of the two gravitational potentials of the conformal Newtonian gauge (script letter): the potential that light feels	new
$\mathcal{D}$, $\mathcal{T}$, $\mathcal{P}$, $\mathcal{A}$	`d_sum` `three_dpres`	$8\pi G_N a_{\rm sp}^2$ times the perturbation of the density, of the divergence of the momentum density, of the pressure, and of the anisotropic stress of matter and radiation together (script letters; in the notation of Ma and Bertschinger: $\delta\rho$, $(\rho+P)\theta$, δP, $(\rho+P)\sigma$)	new
$\mathcal{E}$, $\mathcal{E}_P$, $\mathcal{E}_m$, $\mathcal{E}_r$, $\mathcal{B}$	`grho_sum` `half_rho_plus_p` `b_trace` `n_bg`	$8\pi G_N a_{\rm sp}^2$ times the density and the pressure of the unperturbed universe, summed over the species ($\mathcal{E}_m$, $\mathcal{E}_r$: the parts of $\mathcal{E}$ from pressureless matter and from radiation); and $\mathcal{B} = H'_e - H_e^2 + \frac{1}{2}(\mathcal{E} + \mathcal{E}_P)$, which vanishes without deflation (script letters)	new
$p_{\rm agent}$	—	the pressure in ordinary space of the agent, that is, of whatever keeps the three rates of deflation constant, in the unperturbed universe (under assumption A2 the agent has no energy density)	new
$\mathcal{Y}_M$, $\mathcal{Y}_T$, $\mathcal{Y}_S$; $\mathcal{Z}$	`momentum_gap` `trace_gap` `shear_gap` `x_gap` `shear_check` `trace_check`	the gaps: what the momentum constraint, the trace equation and the equation of the shear lack when only matter and radiation stand on their right-hand sides; the agent that holds the extra scale factors supplies them. $\mathcal{Z} = h'_{\rm sy}\,\mathcal{B} - \frac{1}{2}a_{\rm sp}\,$gsum$\,(\mathcal{D} - 3\mathcal{P})$ (script letters)	new
$Y_{\rm He}$	—	the mass fraction of helium after nucleosynthesis (with the index He; `Y_He` in the data files)	new

10 The metric of the deflating spacetimes in the author's sign convention

symbol	names in programs and data files	meaning	origin
$g^{(4+4)}_{\mu\nu}$, $g^{(7+1)}_{\mu\nu}$	`gs`	the metrics of the deflating spacetimes in the author's sign convention of November 13, 2026 (space-like components positive, time-like components negative): $g^{(4+4)} = \mathrm{diag}(+1,+1,+1,+1,-1,-b_5^2,-b_6^2,-b_7^2)$ and $g^{(7+1)} = \mathrm{diag}(+1,+1,+1,+1,-1,+b_5^2,+b_6^2,+b_7^2)$, that is $g^{(4+4)}_{tt} = -\mathrm{const}_t^2\, e^{-2\,\mathrm{rate}_t\, x4}$ and $g^{(7+1)}_{tt} = +\mathrm{const}_t^2\, e^{-2\,\mathrm{rate}_t\, x4}$ for $t = 5, 6, 7$. A component of the metric carries the superscript $(4+4)$ or $(7+1)$ and two coordinate indices; it is not a rate (g55) and not a coupling (g_r, g_{VS}). The first revision on the deflating spacetimes wrote $g_{1644d} = g^{(4+4)}$ and $g_{1671d} = -g^{(7+1)}$ (time-like component positive on 7+1); these two symbols are retired	author
$\eta^{(4+4)}$, $\eta^{(7+1)}$	—	the flat metrics of the author's sign convention, with which frame indices are raised and lowered on the deflating spacetimes: $\eta^{(4+4)} = \eta_{1644} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ and $\eta^{(7+1)} = -\eta_{1671} = \mathrm{diag}(+1,+1,+1,+1,-1,+1,+1,+1)$; $g_{\mu\nu}\, e_A{}^{\mu} e_B{}^{\nu} = \eta_{AB}$	author
c_γ	`clifford_sign`	the sign that relates the Dirac matrices to the metric of the author's sign convention, $\gamma^A\gamma^B + \gamma^B\gamma^A = 2\, c_\gamma\, \eta^{AB}$ with $\eta = \eta^{(4+4)}$ or $\eta^{(7+1)}$: $c_\gamma = +1$ on 4+4 and $c_\gamma = -1$ on 7+1. The Dirac matrices themselves are those of the flat-space part of the project, unchanged	new
b_t^{ref}	`s5` `s6` `s7`	the value of the scale factor b_t at the reference time $x4_{\mathrm{ref}}$. A particle that keeps its wavenumber k_t has the proper momentum k_t/b_t, larger than at the reference time by the stretch factor $b_t^{\mathrm{ref}}/b_t = e^{\mathrm{rate}_t (x4 - x4_{\mathrm{ref}})}$ (columns `s5`, `s6`, `s7`)	new
τ_d^0	`tau_ref` `tau_ref_end`	the reading that the deflation clock would have if the rates had kept their initial value, $\tau_d^0 = 3\,\mathrm{g55}\, x4$ in the evolutions with gravity (equal rates), where the clock actually reached is $\tau_d = -3 \ln \mathrm{b}$	new

11 Mass term and self-interaction of the fermion field

symbol	names in programs and data files	meaning	origin
$\mathcal{L}_{\mathrm{self}}$; $\mathcal{L}^{\mathrm{C}}_{\mathrm{self}}$, $\mathcal{L}^{\mathrm{Q}}_{\mathrm{self}}$	—	the self-interaction of the fermion field (the author's definition, document KOHN_SHAM_GRAVITY): $\mathcal{L}_{\mathrm{self}} = -\frac{1}{2}\sum_{r=0}^{8} g_r\, I_r$, a sum of nine quartic invariants. On the 4+4 spacetime the invariants can be formed in two ways that are not equivalent: $\mathcal{L}^{\mathrm{C}}_{\mathrm{self}} = -\frac{1}{2}\sum_r g_r\, I^{\mathrm{C}}_r$ (form C, the definition of the 4+4 model) and $\mathcal{L}^{\mathrm{Q}}_{\mathrm{self}} = -\frac{1}{2}\sum_r g_r\, I^{\mathrm{Q}}_r$ (form Q, the alternative)	K1; the two forms on 4+4 are new
g_r; $g_0, \ldots, g_8$; g_S, g_V, g_A, g_P	`g_0` `g_1` `g_2` `g_3` `g_4` `g_5` `g_6` `g_7` `g_8`	the nine couplings of the self-interaction, one for each rank r (the author's symbols; dimension length^{D-1} in D space dimensions); $g_0 = g_S$ (scalar), $g_1 = g_V$ (vector), $g_7 = -g_A$ (axial vector), $g_8 = -g_P$ (pseudoscalar). Such a coupling carries one digit as subscript; it is not a rate of deflation (g55: two digits, upright, no subscript), not a component of the metric (superscript $(4+4)$ or $(7+1)$, two indices) and not the coupling g_{VS} of the Vorton-Scalar	K1

symbol	names in programs and data files	meaning	origin
K, J (index sets); r, r' (ranks)	`mask` `rank` `J`	K, J: a strictly increasing list $A_1 < \cdots < A_r$ of indices out of $0, \ldots, 7$ (there are 256); $r = \lvert K\rvert$ is its rank, $0 \le r \le 8$; r' is a second rank and l the summation index in the closed form of the Fierz matrix. In the data files an index set is given by its bit mask (`mask`: the sum of 2^A over its indices)	K1
Θ^K, Θ_K; ζ_r; sgn_K	`eps_K` `phase` `eta_K`	the 256 matrices of the self-interaction (the author's symbols Θ^K, Θ_K): $\Theta^K = \zeta_r\, \gamma^{A_1} \cdots \gamma^{A_r}$, with the phase $\zeta_r = 1$ or i chosen so that the bilinear formed with Θ^K is real, and $\Theta_K = (\Theta^K)^{-1} = \mathrm{sgn}_K\, \Theta^K$ with the sign $\mathrm{sgn}_K = \pm 1$, so that $(\Theta^K)^2 = \mathrm{sgn}_K$. On 7+1: $\zeta_r = i$ for $r = 2, 3, 6, 7$ (the author's i^{σ_r} with $\sigma_r = r(r-1)/2 \bmod 2$), and sgn_K is the product of the squares of the Dirac matrices in K (the author's η_K; column `eta_K`). On 4+4 (form C, the definition): $\zeta_r = i$ for $r = 1, 2, 5, 6$, and sgn_K is $(-1)^r$ times that product. (The sign is written with the letters sgn, because η and ε have other meanings; columns `eps_K`, `phase`.)	K1; form C of 4+4 is new
γ^A_{44q}; Θ^K_Q, ζ^Q_r, sgn^Q_K	`gamma44q`	the matrices with which form Q (the quantized-sector form) of the self-interaction on 4+4 is built: $\gamma^A_{44q} = -i\,\gamma^A_{1644}$ for $A = 0, \ldots, 4$ and $\gamma^A_{44q} = \gamma^A_{1644}$ for $A = 5, 6, 7$; they have the squares of the 7+1 matrices, and $\gamma^4_{44q} = \beta_{1644}$. $\Theta^K_Q = \zeta^Q_r\, \gamma^{A_1}_{44q} \cdots \gamma^{A_r}_{44q}$ with $\zeta^Q_r = i$ for $r = 2, 3, 6, 7$ and $\zeta^Q_r = 1$ for the other ranks; $(\Theta^K_Q)^2 = \mathrm{sgn}^Q_K$, the product of the squares of the matrices γ^A_{44q} in K	new
I_r; I^{C}_r, I^{Q}_r	—	the invariant of rank r (the author's symbol): $I_r = \sum_{\lvert K\rvert=r} (\psi^\dagger\, \mathrm{Adj}\, \Theta_K \psi)(\psi^\dagger\, \mathrm{Adj}\, \Theta^K \psi)$ with the adjoint matrix Adj of the model. On 7+1: $I_r = \sum_{\lvert K\rvert=r} \mathrm{sgn}_K (\bar\psi\, \Theta^K \psi)^2$. On 4+4 there are two forms: $I^{\mathrm{C}}_r = \sum_{\lvert K\rvert=r} \mathrm{sgn}_K (\tilde\psi\, \Theta^K \psi)^2$ (form C, the definition) and $I^{\mathrm{Q}}_r = \sum_{\lvert K\rvert=r} \mathrm{sgn}^Q_K (\bar\psi\, \Theta^K_Q \psi)^2$ (form Q, the alternative); exactly, $I^{\mathrm{C}}_r = \sum_{\lvert K\rvert=r} \mathrm{sgn}^Q_K (\psi^\dagger W_{1644}\, \beta\, \Theta^K_Q \psi)^2$. I_r without superscript is the invariant of the model at hand; in a statement that holds in both forms of 4+4, of either form. I_0 is the square of the scalar density, I_1 the square of the current	K1; the two forms on 4+4 are new
F, $\mathsf{F}_{rr'}$	`fierz` `F_r0` `F_r1` `F_r2` `F_r3` `F_r4` `F_r5` `F_r6` `F_r7` `F_r8`	the Fierz matrix (9×9): $\mathsf{F}_{rr'} = \frac{1}{16} \sum_l (-1)^{rr'-l} \binom{r'}{l} \binom{8-r'}{r-l}$, with $\mathsf{F}^2 = 1$. For anticommuting fields the nine invariants obey $(1 + \mathsf{F})\, I = 0$. (Sans-serif type; F_S, F_P, F_0, F_1, F_7, F_8 in italic type are pairing amplitudes. The author's document writes F_{rs}.)	K1
g_0^{eff}, g_1^{eff}, g_7^{eff}, g_8^{eff}	`g0eff` `g1eff` `g7eff` `g8eff` `g_0eff` `g_1eff` `g_7eff` `g_8eff`	the four independent couplings that remain for anticommuting fields, when $I_2, \ldots, I_6$ are expressed by I_0, I_1, I_7, I_8: $g_0^{\mathrm{eff}} = g_0 - 3g_2 - 4g_3 - 5g_4 - 4g_5 - g_6$, $g_1^{\mathrm{eff}} = g_1 - g_2 - g_5 + g_6$, $g_7^{\mathrm{eff}} = g_7 - g_2 - g_3 + g_6$, $g_8^{\mathrm{eff}} = g_8 - g_2 + 4g_3 - 5g_4 + 4g_5 - 3g_6$; then $\mathcal{L}_{\mathrm{self}} = -\frac{1}{2}(g_0^{\mathrm{eff}} I_0 + g_1^{\mathrm{eff}} I_1 + g_7^{\mathrm{eff}} I_7 + g_8^{\mathrm{eff}} I_8)$	new
n	—	the one-body density matrix of matter (16×16), $\mathsf{n}_{ab} = \langle \psi^\dagger_b \psi_a \rangle$, with $\mathrm{Tr}\, \mathsf{n} = n$ and $\mathrm{Tr}(\beta\, \mathsf{n}) = n_S$; for uniform matter at rest $\mathsf{n} = (n + n_S\, \beta)/16$. (The author's document writes R for the matrix $\mathsf{n}\, \beta$.)	new

symbol	names in programs and data files	meaning	origin
G_S^{HF}, G_V^{HF}; G_S^{c}, G_V^{c}	`G_0` `G_1` `G_S0` `G_V0` `hf_rho` `hf_rhoS` `hf_rhoW` `hf_rhoSW`	the four combinations of the couplings that the Hartree-Fock approximation gives for uniform matter: $G_S^{\mathrm{HF}} = (15g_0^{\mathrm{eff}} - 8g_1^{\mathrm{eff}} - 8g_7^{\mathrm{eff}} - g_8^{\mathrm{eff}})/16$, $G_V^{\mathrm{HF}} = (-g_0^{\mathrm{eff}} + 22g_1^{\mathrm{eff}} - 6g_7^{\mathrm{eff}} + g_8^{\mathrm{eff}})/16$, $G_S^{\mathrm{c}} = (-g_0^{\mathrm{eff}} - 2g_1^{\mathrm{eff}} + 2g_7^{\mathrm{eff}} + g_8^{\mathrm{eff}})/16$ and $G_V^{\mathrm{c}} = -(g_0^{\mathrm{eff}} + g_8^{\mathrm{eff}})/16$. On 7+1 the interaction energy density is $\frac{1}{2}(G_V^{\mathrm{HF}} n^2 + G_S^{\mathrm{HF}} n_S^2)$: there G_S^{HF}, G_V^{HF} are the mean-field couplings G_S, G_V. On 4+4 it is $\frac{1}{2}(G_V^{\mathrm{tot}} n^2 + G_S^{\mathrm{tot}} n_S^2 + G_V^{\mathrm{dif}} n_W^2 + G_S^{\mathrm{dif}} n_{SW}^2)$ with the same four combinations: $G^{\mathrm{tot}} = G^{\mathrm{c}}$, $G^{\mathrm{dif}} = G^{\mathrm{HF}}$ in form C and $G^{\mathrm{tot}} = G^{\mathrm{HF}}$, $G^{\mathrm{dif}} = G^{\mathrm{c}}$ in form Q. (The superscript c is upright; G_c with the subscript c is the critical coupling of the vacuum. The author's document and the data files write G_0, G_1 for G_S^{HF}, G_V^{HF}; the data files write `G_S0`, `G_V0` for G_S^{c}, G_V^{c}.)	K1; the combinations of 4+4 are new
G_{pair}	`G_pair`	the contact coupling of a pair of two particles at rest: the interaction energy of the pair times the volume; attractive if negative. For the seven vector pairs on 7+1 and in form Q of 4+4: $G_{\mathrm{pair}} = g_0^{\mathrm{eff}} + g_1^{\mathrm{eff}} + 5\, g_7^{\mathrm{eff}}$. In form C of 4+4: $-(g_0^{\mathrm{eff}} + g_1^{\mathrm{eff}} + 5\, g_7^{\mathrm{eff}})$ for the four vector pairs along $x0, x1, x2, x3$ (one particle of each half of the field) and $+(g_0^{\mathrm{eff}} + g_1^{\mathrm{eff}} + 5\, g_7^{\mathrm{eff}})$ for the three vector pairs along $x5, x6, x7$ (two particles of the same half)	new
n_+, n_-, n_{S+}, n_{S-}; n_W, n_{SW}; pol_W	`rho_W` `rho_SW` `pol_W` `pol`	number densities $n_\pm = \langle \psi_\pm^\dagger \psi_\pm \rangle$ and scalar densities $n_{S\pm} = \langle \psi_\pm^\dagger \beta\, \psi_\pm \rangle$ of the particles in the two halves $W_{1644} = \pm 1$ of the 4+4 field; $n = n_+ + n_-$, $n_S = n_{S+} + n_{S-}$, $n_W = n_+ - n_-$, $n_{SW} = n_{S+} - n_{S-}$; the polarization $\mathrm{pol}_W = (n_+ - n_-)/n$ (columns `pol_W`, `pol`)	new
Adj	—	the adjoint matrix of a form of the self-interaction: its bilinears are $\psi^\dagger \mathrm{Adj}\, \Theta^K \psi$. Adj $= \beta$ on 7+1 and in form Q of 4+4, where $\psi^\dagger \mathrm{Adj} = \bar{\psi}$; Adj $= C^-_{1644}$ in form C of 4+4, where $\psi^\dagger \mathrm{Adj} = \tilde{\psi}$. (Upright letters; used where a statement holds for every form.)	new
ψ_+, ψ_-	—	the two halves of the 4+4 field: $\psi = \psi_+ + \psi_-$ with $W_{1644}\, \psi_\pm = \pm \psi_\pm$, eight components each. The Hamiltonian of the quantized sector acts inside each half, and a particle has four internal states in each	new
$\mathcal{L}_{\mathrm{cov}}$	—	the covariant Lagrangian of the free field on 4+4, $\mathcal{L}_{\mathrm{cov}} = \tilde{\psi}\, (\gamma^A \partial_A - m)\, \psi$. For fields of the quantized sector $\mathcal{L}_{\mathrm{cov}} = -\psi^\dagger W_{1644}\, (i\partial_4 - \mathsf{h})\, \psi$: it is $\mathcal{L}_{\mathrm{kin}} + \mathcal{L}_{\mathrm{mass}}$ on the half $W_{1644} = -1$ and minus that on the half $W_{1644} = +1$	new
z_K	—	the phase in the exact relation between the matrices of the two forms on 4+4, $C^-_{1644}\, \Theta^K = z_K\, W_{1644}\, \beta\, \Theta^K_Q$: $z_K = \pm 1$ if the index set K contains an even number of the indices 5, 6, 7, and $z_K = \pm i$ if it contains an odd number	new
$\mathcal{J}_K^{++}$, $\mathcal{J}_K^{--}$, $\mathcal{J}_K^{+-}$, $\mathcal{J}_K^{-+}$	—	the bilinears of the two halves of the 4+4 field for the index set K: $\mathcal{J}_K^{++} = \psi_+^\dagger \beta\, \Theta^K_Q \psi_+$, $\mathcal{J}_K^{--} = \psi_-^\dagger \beta\, \Theta^K_Q \psi_-$, $\mathcal{J}_K^{+-} = \psi_+^\dagger \beta\, \Theta^K_Q \psi_-$ and $\mathcal{J}_K^{-+} = \psi_-^\dagger \beta\, \Theta^K_Q \psi_+ = (\mathcal{J}_K^{+-})^\dagger$. If K contains an even number of the indices 5, 6, 7 only the first two differ from zero, otherwise only the last two. (Script letter with an index set and two signs; J^A_{VS}, J^A_5 are currents and J is an index set.)	new

symbol	names in programs and data files	meaning	origin
$G_S^{\rm tot}$, $G_V^{\rm tot}$; $G_S^{\rm dif}$, $G_V^{\rm dif}$	`G_Stot` `G_Vtot` `G_Sdif` `G_Vdif`	the mean-field couplings of the two halves of the 4+4 field: the interaction energy density of uniform matter is $\frac{1}{2}(G_V^{\rm tot}n^2 + G_S^{\rm tot}n_S^2 + G_V^{\rm dif}n_W^2 + G_S^{\rm dif}n_{SW}^2)$. $G^{\rm tot}$ multiply the squares of the total densities, $G^{\rm dif}$ the squares of the differences between the densities of the two halves. Form C: $G_S^{\rm tot} = G_S^{\rm c}$, $G_V^{\rm tot} = G_V^{\rm c}$, $G_S^{\rm dif} = G_S^{\rm HF}$, $G_V^{\rm dif} = G_V^{\rm HF}$. Form Q: $G_S^{\rm tot} = G_S^{\rm HF}$, $G_V^{\rm tot} = G_V^{\rm HF}$, $G_S^{\rm dif} = G_S^{\rm c}$, $G_V^{\rm dif} = G_V^{\rm c}$. (In the data files and in the statements of the programs: `G_Stot`, `G_Vtot`, `G_Sdif`, `G_Vdif`.)	new
$G_S^{\rm same}$, $G_V^{\rm same}$; $G_S^{\rm opp}$, $G_V^{\rm opp}$	`G_Ssame` `G_Vsame` `G_Sopp` `G_Vopp`	the mean-field couplings between two particles of the same half and between particles of opposite halves of the 4+4 field: $G^{\rm same} = G^{\rm tot} + G^{\rm dif}$ and $G^{\rm opp} = G^{\rm tot} - G^{\rm dif}$, so that the interaction energy density is $\frac{1}{2}G_V^{\rm same}(n_+^2 + n_-^2) + G_V^{\rm opp}\, n_+ n_- + \frac{1}{2}G_S^{\rm same}(n_{S+}^2 + n_{S-}^2) + G_S^{\rm opp}\, n_{S+}n_{S-}$. $G^{\rm same} = G^{\rm HF} + G^{\rm c}$ is the same in both forms: $G_S^{\rm same} = (7g_0^{\rm eff} - 5g_1^{\rm eff} - 3g_7^{\rm eff})/8$ and $G_V^{\rm same} = (-g_0^{\rm eff} + 11g_1^{\rm eff} - 3g_7^{\rm eff})/8$. $G^{\rm opp}$ has opposite signs in the two forms: $G^{\rm opp} = G^{\rm HF} - G^{\rm c}$ in form Q and $G^{\rm opp} = -(G^{\rm HF} - G^{\rm c})$ in form C, with $G_S^{\rm HF} - G_S^{\rm c} = (8g_0^{\rm eff} - 3g_1^{\rm eff} - 5g_7^{\rm eff} - g_8^{\rm eff})/8$ and $G_V^{\rm HF} - G_V^{\rm c} = (11g_1^{\rm eff} - 3g_7^{\rm eff} + g_8^{\rm eff})/8$. (In the data files and in the statements of the programs: `G_Ssame`, `G_Vsame`, `G_Sopp`, `G_Vopp`.)	new
g (strength)	—	the common strength of the couplings of the Lagrangian in the sets A and B of the numerical examples on 4+4: $g = \|g_0\| > 0$, of dimension length^{D-1} (column `g` of the files `forms_..`, in units $m = 1$; in the files of the gaps the column `g` is the coupling g_{VS}). On the deflating spacetime g grows as e^{τ_d}	new
$\mathcal{L}_{\rm kin}$, $\mathcal{L}_{\rm mass}$	—	the kinetic term and the mass term of the Lagrangian of the fermion field, $\mathcal{L} = V_d\,[\mathcal{L}_{\rm kin} + \mathcal{L}_{\rm mass} + \mathcal{L}_{\rm self}]$. On 4+4, for fields of the quantized sector, $\mathcal{L}_{\rm kin} + \mathcal{L}_{\rm mass} = \psi^\dagger(i\partial_4 - \mathsf{h})\psi$ (the positive Lagrangian, with the mass term $-m\,\bar\psi\psi$)	new

12 Kohn-Sham and CJT calculations with contact interaction

symbol	names in programs and data files	meaning	origin
G_S, G_V	`G_S` `G_V` `GS_abs`	the scalar and the vector coupling of the Kohn-Sham (mean-field) description of uniform matter (the author's symbols): interaction energy density $\frac{1}{2}G_V n^2 + \frac{1}{2}G_S n_S^2$; attraction for $G_S < 0$, repulsion for $G_V > 0$; dimension length^{D-1}. The author's example: $G_S = -10^5$, $G_V = +10^4$ in units $m = 1$, $D = 7$	K1
n, n_S	`rho` `rho_S` `rho_ks` `rho_cjt` `n_ks` `n_cjt`	number density $\langle\psi^\dagger\psi\rangle$ and scalar density $\langle\bar\psi\psi\rangle$ of the fermions of matter (the filled states of negative energy not counted). The data files and the author's document write ρ (`rho`) and ρ_S (`rho_S`); the files of the two halves of the 4+4 field write `n`	K1 (there ρ, ρ_S)
D	`D`	the number of space dimensions in which the particles of matter have momentum: 7 (all space coordinates of 7+1), 4 (fields that do not depend on $x5, x6, x7$) or 3 (fields that depend on ordinary space only). $V_D = \pi^{D/2}/\Gamma_{\rm E}(D/2+1)$ is the volume and $A_D = D\,V_D$ the area of the unit ball in D dimensions ($\Gamma_{\rm E}$: Euler's gamma function)	new

symbol	names in programs and data files	meaning	origin
$E^*(k)$; $\langle f\rangle$	—	$E^*(k)=\sqrt{k^2+m^{*2}}$, the energy of a particle of matter without the vector term; $\langle f\rangle=\nu\int d^Dk\, f/(2\pi)^D$ over the occupied states: $n=\langle 1\rangle$, $n_S=\langle m^*/E^*\rangle$. For the two halves of the 4+4 field: $E^*_\pm(k)=\sqrt{k^2+m^{*2}_\pm}$	K1 (there ω for E^*)
$\varepsilon_{\rm kin}$; p_j; $p_{\rm int}$	`e_kin` `eps` `eps_ks` `eps_cjt` `eps_rel` `p` `p_ks` `p_flat` `p5` `p6` `p7` `p_a` `p_b` `pa_rel` `pb_rel` `Pi`	$\varepsilon_{\rm kin}=\langle E^*\rangle$: the kinetic part of the energy density $\varepsilon=\varepsilon_{\rm kin}+\frac12 G_V n^2-(m^*-m)^2/(2G_S)$; p_j: the pressure of matter along the coordinate xj, $\langle k_j^2/E^*\rangle+p_{\rm int}$; $p_{\rm int}=\frac12 G_V n^2+(m^*-m)^2/(2G_S)$: the interaction part of the pressure, which is the whole pressure along a coordinate on which the fields do not depend (`Pi` in the data files, Π in the author's document)	K1
V_0	`V_0`	the vector mean field $V_0=G_V\,n$ (the author's symbol); it shifts the chemical potential, $\mu\to\mu-V_0$. For the two halves of the 4+4 field: $V_{0\pm}=G_V^{\rm tot}\,n\pm G_V^{\rm dif}\,n_W$	K1
$\Omega_{\rm CJT}$; $\Omega_{\rm half}$	`minus_Omega`	the CJT effective potential of uniform matter with contact interaction per unit volume, a function of the mass m^* and of the vector field V_0 at the chemical potential μ: $\Omega_{\rm CJT}=-\nu\int\frac{d^Dk}{(2\pi)^D}\,(\mu-V_0-E^*)\,\theta(\mu-V_0-E^*)-\frac{(m^*-m)^2}{2G_S}-\frac{V_0^2}{2G_V}$; at its stationary point $-\Omega_{\rm CJT}$ is the pressure. For the two halves of the 4+4 field it is a function of two masses and two vector fields: $\Omega_{\rm CJT}=\Omega_{\rm half}(m^*_+,\mu-V_{0+})+\Omega_{\rm half}(m^*_-,\mu-V_{0-})$ $-[\frac12(m^*_++m^*_-)-m]^2/(2G_S^{\rm tot})-[\frac12(m^*_+-m^*_-)]^2/(2G_S^{\rm dif})$ $-[\frac12(V_{0+}+V_{0-})]^2/(2G_V^{\rm tot})-[\frac12(V_{0+}-V_{0-})]^2/(2G_V^{\rm dif})$, with the grand potential of free particles of one half (four internal states), $\Omega_{\rm half}(m^*,\mu)=-4\int\frac{d^Dk}{(2\pi)^D}\,(\mu-E^*)\,\theta(\mu-E^*)$	new
G_c	`Gc`	the critical coupling of the vacuum: with the filled states of negative energy below the cutoff Λ a massless field acquires a mass when $G_S<G_c=-(2\pi)^D(D-1)/(\nu A_D\Lambda^{D-1})$; $G_c\Lambda^6=-90\pi^4$ for $D=7$	K2 ($D=7$)
$m_{\rm vac}$, $m_{\rm bare}$; $\langle\bar\psi\psi\rangle_{\rm vac}$	`M_vac` `m_bare` `condensate`	with the filled states of negative energy: the mass of the fermions in vacuum, the mass parameter of the Lagrangian (bare mass) that produces it, and the scalar density of the filled states, $\langle\bar\psi\psi\rangle_{\rm vac}=-\nu\int^\Lambda\frac{d^Dk}{(2\pi)^D}\frac{m_{\rm vac}}{\sqrt{k^2+m_{\rm vac}^2}}$ (negative); $m_{\rm vac}=m_{\rm bare}+G_S\langle\bar\psi\psi\rangle_{\rm vac}$	new
$n^{\rm vac}_{S+}$, $n^{\rm vac}_{S-}$; $\Omega_{\rm vac}$	`Omega`	for the two halves of the 4+4 field with the filled states of negative energy below the cutoff Λ: the scalar densities of the filled states, $n^{\rm vac}_{S\pm}=-4\int_{\vec k<\Lambda}\frac{d^Dk}{(2\pi)^D}\,\frac{m^*_\pm}{E^*_\pm}$, which enter the gap equations of the vacuum $\frac12(m^*_++m^*_-)-m=G_S^{\rm tot}(n^{\rm vac}_{S+}+n^{\rm vac}_{S-})$, $\frac12(m^*_+-m^*_-)=G_S^{\rm dif}(n^{\rm vac}_{S+}-n^{\rm vac}_{S-})$; and the energy density of the vacuum relative to the state in which both halves have the mass m of the Lagrangian, $\Omega_{\rm vac}=-4\sum_\pm\int_{\vec k<\Lambda}\frac{d^Dk}{(2\pi)^D}\,[E^*_\pm-\sqrt{k^2+m^2}]$ $-[\frac12(m^*_++m^*_-)-m]^2/(2G_S^{\rm tot})-[\frac12(m^*_+-m^*_-)]^2/(2G_S^{\rm dif})$; among several solutions of the gap equations the one of lowest $\Omega_{\rm vac}$ is the vacuum (column `Omega`)	new
$g_{\rm crit}$; $\Lambda_{\rm max}$	`g_crit` `cutoff_max`	$g_{\rm crit}$: the critical strength of a set of couplings, the strength g at which the more attractive of the scalar couplings $G_S^{\rm tot}$, $G_S^{\rm dif}$ reaches the critical coupling G_c of the vacuum of a massless field. $\Lambda_{\rm max}$: at a given strength, the largest cutoff Λ for which that coupling is still weaker than G_c, so that the vacuum of the massless field is stable (column `cutoff_max`, in units of m)	new

symbol	names in programs and data files	meaning	origin
k_{F+}, k_{F-}, $k_{F,\mathrm{tot}}$; μ_+, μ_-	`kF_total` `mu_plus` `mu_minus`	for the two halves of the 4+4 field: the Fermi momenta of the two halves, $n_\pm = 4V_D k_{F\pm}^D/(2\pi)^D$ (four internal states each); the Fermi momentum $k_{F,\mathrm{tot}}$ that the total density would have with equal halves, $n = 8V_D k_{F,\mathrm{tot}}^D/(2\pi)^D$; and the chemical potentials $\mu_\pm = \sqrt{k_{F\pm}^2 + m_\pm^{*2}} + V_{0\pm}$ of the two halves (for an empty half: the energy of its lowest state)	new
$\mathsf{R}_{\mathrm{gap}}$	`response`	the response matrix of the two gap equations $m_\pm^* = m + G_S^{\mathrm{same}} n_{S\pm} + G_S^{\mathrm{opp}} n_{S\mp}$ of the halves of the 4+4 field: the 2×2 matrix with the elements $(\mathsf{R}_{\mathrm{gap}})_{++} = G_S^{\mathrm{same}}\, \partial n_{S+}/\partial m_+^*$, $(\mathsf{R}_{\mathrm{gap}})_{+-} = G_S^{\mathrm{opp}}\, \partial n_{S-}/\partial m_-^*$, $(\mathsf{R}_{\mathrm{gap}})_{-+} = G_S^{\mathrm{opp}}\, \partial n_{S+}/\partial m_+^*$, $(\mathsf{R}_{\mathrm{gap}})_{--} = G_S^{\mathrm{same}}\, \partial n_{S-}/\partial m_-^*$ (derivatives at fixed Fermi momenta). While its eigenvalues are below 1 the solution is the regular continuation of the dilute gas; at the eigenvalue 1 solutions with other masses branch off. Column `response`: the largest eigenvalue. (Sans-serif with the suffix gap; R in italic type is a curvature.)	new
$-g_0/C$, g_1/C	`kappa_0` `kappa_1`	the couplings g_0 and g_1 of the Lagrangian in units of the coupling C of the Vorton-Scalar (units $m = 1$), in the calculations with bare mass, self-interaction and Vorton-Scalar together on 4+4; they are written as these ratios, without a letter of their own (columns `kappa_0`, `kappa_1`)	new
m_D, m_Y, f_D	`f_D`	when a bare Dirac mass and the Vorton-Scalar are both present: the bare Dirac mass m_D, the part $m_Y = g_{VS} v_{VS}$ of the mass that the Vorton-Scalar gives, and the fraction $f_D = m_D/m$ of the vacuum mass $m = m_D + m_Y$	new
m_ϑ	`mph2_over_C_ms2`	the mass of the phase mode ϑ_{VS} of the Vorton-Scalar: zero while the chiral symmetry $U(1)_{VS}$ is exact; a bare Dirac mass breaks the symmetry and gives $m_\vartheta^2 v_{VS}^2 = (\lvert\langle\bar\psi\psi\rangle_{\mathrm{vac}}\rvert - n_S) m_D m_Y/m$	new
$-G_S/C$, G_V/C	`kappa_S` `kappa_V`	the contact couplings in units of the coupling $C = g_{VS}^2 m^2/m_s^2$ of the Vorton-Scalar (units $m = 1$); they are written as these ratios, without a letter of their own (columns `kappa_S`, `kappa_V`)	new
h_{KS}	—	the Hamiltonian matrix of a particle of matter in the Kohn-Sham description, $\mathsf{h}_{\mathrm{KS}} = \sum_j \alpha^j k_j + \beta\, m^* + V_0$	K1
$\Gamma[S]$; S, S_0	—	the CJT effective action as a functional of the propagator of the fermions, $\Gamma[S] = -i\,\mathrm{Tr}\ln S^{-1} - i\,\mathrm{Tr}(S_0^{-1}S - 1) + \Gamma_2[S]$, with the full propagator S and the propagator S_0 of free particles of the mass m (the argument in brackets tells it from the action $\Gamma[\phi, G]$ of the model with the Vorton-Scalar)	new
n_{ref}	—	the number density of matter at the reference time $x4_{\mathrm{ref}}$; matter that keeps its particles has $n = n_{\mathrm{ref}}\, e^{\tau_d}$	new

13 Pairing with contact couplings

symbol	names in programs and data files	meaning	origin
F_0, F_1, F_7, F_8	`F0` `F1` `F7` `F8` `F0c` `F1c` `F7c` `F8c`	the pairing amplitudes of the four independent contact channels in the pair channel along the hidden axis, for particles of mass m^* that scatter from the momenta $\vec{p}, -\vec{p}$ to $\vec{p}\,', -\vec{p}\,'$ in ordinary space ($E = \sqrt{m^{*2} + p^2}$, E' likewise): $F_0 = (EE' + m^{*2} - \vec{p}\cdot\vec{p}\,')/(2EE')$, $F_1 = (3EE' - 2m^{*2} - \vec{p}\cdot\vec{p}\,')/(EE')$, $F_7 = (3EE' + 2m^{*2} - \vec{p}\cdot\vec{p}\,')/(EE')$, $F_8 = -(EE' - m^{*2} - \vec{p}\cdot\vec{p}\,')/(2EE')$. $F_0 = F_S$ and $F_8 = -F_P$. The same closed forms hold in the pair channels along $x5, x6, x7$; the sign with which they enter the pairing potential is $s_{\rm ct}$	new
$m_{\rm ct}$	`m_ct`	the range parameter of the contact couplings in the gap equation: each effective coupling $g_r^{\rm eff}$ is replaced by $g_r^{\rm eff}\, m_{\rm ct}^2/(q^2 + m_{\rm ct}^2)$, the form that the exchange of a boson of mass $m_{\rm ct}$ gives (q: momentum transfer)	new
$V_{\rm ct}$, V_{VS}	—	the pairing potentials in the pair channel along the hidden axis and in those along $x5, x6, x7$: of the contact interaction, $V_{\rm ct} = s_{\rm ct}\sum_{r=0,1,7,8} g_r^{\rm eff} F_r\, m_{\rm ct}^2/(q^2 + m_{\rm ct}^2)$, and of the exchange of the two modes of the Vorton-Scalar, $V_{VS} = -g_{VS}^2[F_S/(q^2 + m_s^2) + F_P/(q^2 + m_\vartheta^2)]$; attractive where negative	new
$s_{\rm ct}$	`ct_sign`	the sign with which the contact couplings enter the pairing potential $V_{\rm ct}$ of a vector pair whose axis is perpendicular to ordinary space: $s_{\rm ct} = +1$ on 7+1, in form Q of 4+4 (the pair along the hidden axis $x0$ and the pairs along $x5, x6, x7$) and in form C for the pairs along $x5, x6, x7$; $s_{\rm ct} = -1$ in form C for the pair along the hidden axis (column `ct_sign`)	new
$k_{\rm max}$	`K_MAX` `K_MAX_EARLIER`	the upper limit of the momentum integral of the gap equation. In the calculations of this folder $k_{\rm max} = 2\times 10^4 \max(m, k_F)$ (constant `K_MAX` of the program); the earlier calculations of the project used $60\max(m, k_F)$ (`K_MAX_EARLIER`)	new
$w_\infty(t)$; s_Δ; $U_{\rm uv}$; u_S, u_V	`uv` `UV_I_S` `UV_I_V` `UV_CRITICAL`	the quantities of the ultraviolet criterion of the gap equation. $w_\infty(t)$: the kernel at momenta far above all masses, where $W(k,k') \to w_\infty(k'/k)/k^2$. s_Δ: the exponent of a power-law solution $\Delta(k) \propto k^{-s_\Delta}$ at large momenta, a root of $\int_0^\infty dt\, t^{1-s_\Delta}\, w_\infty(t) = 4\pi^2$. $U_{\rm uv} = \int_0^\infty dt\, w_\infty(t)$: the ultraviolet strength of the kernel; the gap equation has a solution that does not depend on $k_{\rm max}$ if and only if $U_{\rm uv} < 4\pi^2$. With both modes of the Vorton-Scalar, $U_{\rm uv} = 2u_S\, g_{VS}^2 - s_{\rm ct}\, m_{\rm ct}^2[(g_0^{\rm eff} - g_8^{\rm eff})\, u_S + (g_1^{\rm eff} + g_7^{\rm eff})\, u_V]$, with the two integrals $u_S = \pi^2/8 - \frac{1}{2}$ and $u_V = 3\pi^2/4 - 1$ (constants `UV_I_S`, `UV_I_V`; `UV_CRITICAL` is $4\pi^2$). Column `uv`: $U_{\rm uv}/4\pi^2$	new
Φ, $\Phi(\vec{p})$	—	a state of a pair of two particles, an antisymmetric 16×16 matrix; $\Phi(\vec{p})$: the state of the pair along the hidden axis with the momenta $\vec{p}, -\vec{p}$, the matrix $(C^-\gamma^0)^*$ between the projectors on the states of positive energy. On 4+4: $W_{1644}\,\Phi\, W_{1644}^T = -\Phi$ for a pair of one particle of each half of the field and $+\Phi$ for a pair of two particles of the same half. (Without suffix; the Vorton-Scalar always carries the suffix VS.)	new

14 Einstein-Lovelock gravity with Kohn-Sham matter

symbol	names in programs and data files	meaning	origin
$E^{(k)\mu}{}_{\nu}$	—	the Lovelock tensor of order $k = 1, 2, 3$ in the author's notation and sign: $E^{(1)} = -\mathcal{G}$ (minus the Einstein tensor), $E^{(2)} = \mathcal{H}$ (the Lanczos tensor). For a metric whose scale factors depend only on $x4$: $E^{(k)m}{}_{m} = (-1)^{k+1} k!\, 2^{k-1}$ times the sum, over the sets of $2k$ frame directions other than m and over their divisions into pairs, of the products of the sectional curvatures K_{AB} of the pairs. (With the order in parentheses as a superscript; E_F, E_k, $E_\perp$, E^* are energies.)	K1
α_1, α_2, α_3; $\bar{\Lambda}$; κ_8	`LambdaBar` `kappa` `kappa_eps`	the couplings of the Lovelock terms ($\alpha_1 = 1$), the cosmological term and the gravitational coupling of the author's Einstein-Lovelock equations: $\sum_k \alpha_k E^{(k)4}{}_{4} - \bar{\Lambda} = \kappa_8\, \varepsilon$ for the time component and $\sum_k \alpha_k E^{(k)m}{}_{m} - \bar{\Lambda} = -\kappa_8\, p_m$ for the space direction m. $\alpha_2 = -\alpha_{GB}$ and $\bar{\Lambda} = \Lambda_8$. The author's document writes κ for $\kappa_8 = 8\pi G_8$ (the letter κ with an argument, $\kappa(k)$, is a pair correlation). A Lovelock coupling carries a digit as subscript; α^j with a superscript are Dirac matrices	K1
a, b; H_{a}, H_{b}; n_{a}, n_{b}	`a` `b` `H_a` `H_b` `Ha_over_g` `Hb_over_g` `Hb0_over_g` `a_end` `b_end` `Ha_end_over_g` `Hb_end_over_g`	the two scale factors of a spacetime whose space coordinates form two groups of n_{a} and n_{b} coordinates, and their rates $H_{\mathsf{a}} = \mathsf{a}'/\mathsf{a}$, $H_{\mathsf{b}} = \mathsf{b}'/\mathsf{b}$ (a prime: derivative with respect to the time $x4$). In the author's document: a for ordinary space ($n_{\mathsf{a}} = 3$) and b for the four auxiliary coordinates. On the deflating spacetimes with equal rates: a for $x0, \ldots, x3$ ($n_{\mathsf{a}} = 4$) and b $= b_5 = b_6 = b_7$ ($n_{\mathsf{b}} = 3$); the undisturbed deflation is $H_{\mathsf{a}} = 0$, $H_{\mathsf{b}} = -\mathsf{g55}$	K1 (there a, b, H_a, H_b)
loading	`loading`	$\kappa_8\, \varepsilon(0)/(3\, \mathsf{g55}^2)$: the energy density of the matter at the start of an evolution, in units of the energy density that would make three coordinates deflate at the rate g55 in Einstein gravity	new
$T^{\mu}{}_{\nu}$; p_{a}, p_{b}; δH_{b}	—	the energy-momentum tensor of uniform matter in mixed components, $T^4{}_4 = \varepsilon$, $T^m{}_m = -p_m$; its pressures along the coordinates that carry the scale factors a and b; the deviation $H_{\mathsf{b}} + \mathsf{g55}$ of the rate of the deflating coordinates from the undisturbed deflation	new

15 Tags and identifiers of the calculations with mass term and self-interaction

symbol	names in programs and data files	meaning	origin
71s, 44q, 44c	`71s` `44q` `44c`	tags of the three models of the self-interaction in the programs and the data files: 71s on 7+1; 44c form C on 4+4, the definition (bilinears with $\tilde{\psi} = \psi^\dagger C^-_{1644}$); 44q form Q on 4+4, the alternative (bilinears with $\bar{\psi} = \psi^\dagger \beta_{1644}$ and the matrices γ^A_{44q})	new

symbol	names in programs and data files	meaning	origin
form C, form Q	—	the two forms of the self-interaction on the 4+4 spacetime. Form C, the covariant form, is the definition of the 4+4 model (instruction of the author of November 13, 2026): the invariants I_r^{C}, built with the adjoint $\tilde{\psi}$, which are invariant under all rotations and boosts of the 4+4 frame, together with the positive kinetic and mass terms. Form Q, the quantized-sector form, is the alternative: the invariants I_r^{Q}, built with the adjoint $\bar{\psi}$ of the positive kinetic term. In the data files the column `form` holds C or Q, or 0 for C and 1 for Q. (Upright capitals; C in italic type is the coupling of the Vorton-Scalar, and $C^{\pm}$ with a sign are charge-conjugation matrices.)	new
set A, set B	—	the two sets of couplings of the Lagrangian for which the numbers of the two forms on 4+4 are given, each with one strength g: set A, a scalar attraction alone, $g_0 = -g$; set B, $g_0 = -g$ and $g_1 = +0.1\,g$ (the ratio of the author's example); all other couplings zero. In the data files of the two forms the column `set` holds A or B, or 0 for A and 1 for B (in the files of the gaps the column `set` numbers the pairs of values of m_s and k_F)	new
identifiers of the machine checks	—	`44d-..`, `71d-..`, `def-..`: checks on the deflating 4+4 spacetime, on the deflating 7+1 spacetime, and of results that hold on both; `44-..`, `71-..`: checks of the flat spacetimes, executed again; `71s-S..`, `44q-S..`, `44c-S..`, `self-X..`: the algebra of the self-interaction; `ks-K..`: Kohn-Sham calculations; `cjt-J..`: CJT effective potential with contact interaction; `uni-U..`: contact interaction, bare mass and Vorton-Scalar together; `pc-P..`: pairing with contact couplings; `gr-G..`: gravity with Kohn-Sham matter; `fm-F..`: the two forms of the self-interaction on 4+4 (matter of two halves, vacuum, thresholds). A leading capital I (`Icks-..`, `Idef-..`, `I44d-..`, `I71d-..`) marks a check of the independent verification in Python	new

8. Key to the column "origin"

code	source
N01	the author's notebook `4+4.nb` (Dirac matrices of 4+4; split octonions)
N02	the author's notebook `7+1.nb` (octonions; Dirac matrices and Dirac equation of 7+1)
N04	the author's notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb` (meaning of the coordinates)
N05	the author's notebook `Dirac_Nash_Identity.nb`
N10	the author's notebook `Pair_Creation_of_Universes_WaveFunctionOfUniverse-4+4-Einstein-Lovelock-Nash.nb`
N12	the author's notebook `WaveFunctionUniverse-4+4-Einstein-Lovelock.nb`
D06	P. L. Nash, "Identities satisfied by the generators of the Dirac algebra", J. Math. Phys. 25, 204 (1984)
D07	P. L. Nash, "On the structure of the split octonion algebra", Nuovo Cimento B 105, 31 (1990)
D08	P. L. Nash, "Second gravity", J. Math. Phys. 51, 042501 (2010)

code	source
K1	the author's document KOHN_SHAM_GRAVITY (folder 7+1; sections 15 and 16 of the notebook 7+1.nb): nine-channel self-coupling, Fierz matrix, Kohn-Sham theory, Einstein-Lovelock evolution
K2	the author's document DIRAC16COMPLEX (folder 7+1): the critical coupling of the vacuum in seven space dimensions
author	a name that an instruction of the author fixes (the rates g55, g66, g77; the constants; the signs of the metric)
new	introduced by this project